Supporting file

Nomaer Rahman

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```
library(tinytex)
library(ggplot2)
```

1. Organise and clean the data

1.1 Subset the data into the specific dataset allocated

```
SID <-
SIDoffset <- (SID %% 50) + 1
load("carbon-footprint.Rda")
mydf <- df[seq(from=SIDoffset,to=nrow(df),by=50),]
```

1.2 Data quality plan

```
str(mydf)
```

```
## 'data.frame':    240 obs. of  11 variables:
##  $ household_size             : int  2 2 1 2 4 1 1 2 2 2 ...
##  $ annual_energy_consumption  : num  3533 3793 4912 4895 5404 ...
##  $ recycles_regularly         : int  0 0 1 0 0 0 1 0 1 1 ...
##  $ uses_public_transport      : chr  "No" "No" "Yes" "Yes" ...
##  $ diet_type                  : chr  "Vegetarian" "Vegetarian" "Omnivore" "Vegan" ...
##  $ carbon_footprint           : num  10573 10391 10462 13876 13399 ...
##  $ environmental_concern_score: int  4 5 7 5 4 7 5 4 6 8 ...
##  $ income                     : num  35308 29181 11165 36710 13390 ...
##  $ number_adults              : int  2 2 1 2 4 1 1 2 2 2 ...
##  $ home_type                  : chr  "Detached" "Semi-detached" "Semi-detached" "Terraced" ...
##  $ recycles_regularly_question: chr  "No" "No" "Yes" "No" ...
```

```
head(mydf)
```

```
##     household_size annual_energy_consumption recycles_regularly
## 43               2                    3532.6                  0
## 93               2                    3793.2                  0
## 143              1                    4912.4                  1
```


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```
## 193              2                   4894.7                 0
## 243              4                   5403.5                 0
## 293              1                   4653.4                 0
##     uses_public_transport  diet_type carbon_footprint
## 43                     No Vegetarian          10573.2
## 93                     No Vegetarian          10391.4
## 143                   Yes   Omnivore          10461.9
## 193                   Yes      Vegan          13875.5
## 243                    No      Vegan          13398.8
## 293                   Yes      Vegan          10701.8
##     environmental_concern_score   income number_adults     home_type
## 43                            4 35307.57             2      Detached
## 93                            5 29180.57             2 Semi-detached
## 143                           7 11164.67             1 Semi-detached
## 193                           5 36709.70             2      Terraced
## 243                           4 13390.20             4      Terraced
## 293                           7 13128.95             1 Semi-detached
##     recycles_regularly_question
## 43                           No
## 93                           No
## 143                         Yes
## 193                          No
## 243                          No
## 293                          No
```

```
tail(mydf)
```

```
##       household_size annual_energy_consumption recycles_regularly
## 11743              4                    4907.6                  1
## 11793              1                    3616.3                  1
## 11843              3                    5829.2                  1
## 11893              3                    6447.3                  0
## 11943              1                    2622.6                  0
## 11993              1                    4141.6                  0
##       uses_public_transport  diet_type carbon_footprint
## 11743                   Yes   Omnivore          12370.0
## 11793                    No      Vegan           9254.2
## 11843                   Yes   Omnivore          11311.8
## 11893                    No   Omnivore          15000.0
## 11943                    No   Omnivore           5629.5
## 11993                    No Vegetarian           9493.5
##       environmental_concern_score   income number_adults     home_type
## 11743                          10 28051.39             4          Flat
## 11793                           8 14282.75             0 Semi-detached
## 11843                           8  5595.56             3      Detached
## 11893                           6 28492.64             3      Terraced
## 11943                           4 10034.66             1      Terraced
## 11993                           5 17175.38             0      Detached
##       recycles_regularly_question
## 11743                         Yes
## 11793                         Yes
## 11843                         Yes
## 11893                          No
## 11943                          No
```

```
## 11993                          No
```

```
summary(mydf)
```

```
##  household_size  annual_energy_consumption recycles_regularly
##  Min.   :1.000   Min.   :-650              Min.   :0.0000
##  1st Qu.:1.000   1st Qu.:3813              1st Qu.:0.0000
##  Median :2.000   Median :4393              Median :1.0000
##  Mean   :2.404   Mean   :4331              Mean   :0.5292
##  3rd Qu.:3.000   3rd Qu.:5033              3rd Qu.:1.0000
##  Max.   :7.000   Max.   :6942              Max.   :1.0000
##
##  uses_public_transport  diet_type         carbon_footprint
##  Length:240            Length:240         Min.   : 5093
##  Class :character      Class :character   1st Qu.: 9366
##  Mode  :character      Mode  :character   Median :10894
##                                           Mean   :10940
##                                           3rd Qu.:12525
##                                           Max.   :15000
##
##  environmental_concern_score     income       number_adults
##  Min.   : 1.000              Min.   :  5256   Min.   :0.000
##  1st Qu.: 5.000              1st Qu.: 12303   1st Qu.:1.000
##  Median : 6.000              Median : 19040   Median :2.000
##  Mean   : 5.917              Mean   : 22021   Mean   :2.288
##  3rd Qu.: 7.000              3rd Qu.: 27016   3rd Qu.:3.000
##  Max.   :10.000              Max.   :153423   Max.   :7.000
##                              NA's   :22
##   home_type         recycles_regularly_question
##  Length:240         Length:240
##  Class :character   Class :character
##  Mode  :character   Mode  :character
##
##
##
##
```

```
sum(is.na(mydf$income))
```

```
## [1] 22
```

```
mydf[mydf$annual_energy_consumption < 1000 | mydf$annual_energy_consumption > 9000, ]
```

```
##      household_size annual_energy_consumption recycles_regularly
## 3043              1                     -500                  1
## 3093              1                     -500                  0
## 6043              6                     -650                  1
## 6093              2                     -650                  0
##      uses_public_transport diet_type carbon_footprint
## 3043                   Yes  Omnivore          10814.4
## 3093                    No     Vegan           7170.5
## 6043                    No  Omnivore          13117.7
```

```
## 6093                    No  Omnivore           9862.6
##      environmental_concern_score   income number_adults home_type
## 3043                           8       NA             1      Flat
## 3093                           4       NA             1  Detached
## 6043                           4 22353.31             6  Detached
## 6093                           3 12987.12             2      Flat
##      recycles_regularly_question
## 3043                         Yes
## 3093                          No
## 6043                         Yes
## 6093                          No
```

```
unique(mydf$home_type)
```

```
## [1] "Detached"      "Semi-detached" "Terraced"      "Flat"
## [5] "flat"          "Terrrace"      "semi-detached" "detatched"
```

```
sum(mydf$number_adults == 0)
```

```
## [1] 28
```

```
testmydf <- mydf[mydf$number_adults != 0, ]
all(testmydf$household_size == testmydf$number_adults)
```

```
## [1] TRUE
```

```
mydf$uses_public_transport<-as.factor(mydf$uses_public_transport)

sum(is.na((mydf$recycles_regularly == as.factor(mydf$recycles_regularly_question))))
```

```
## [1] 0
```

After reviewing R's interpretation of the data, I examined each data type column by column and found some notable points. I then investigated these areas and identified several key issues: negative values in annual_energy_consumption, missing values in income, misspellings in home_types, and duplicate columns between household_size and number_adults—excluding rows where there is a 0 in number_adults and a household_size of 1, which seems very odd.

1.3 Data quality analysis findings

#Issue 1 : Implausible values in annual_energy_consumption

```
mydf[mydf$annual_energy_consumption < 1000 | mydf$annual_energy_consumption > 9000, ]
```

```
##      household_size annual_energy_consumption recycles_regularly
## 3043              1                     -500                   1
## 3093              1                     -500                   0
## 6043              6                     -650                   1
## 6093              2                     -650                   0
```


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```
##      uses_public_transport diet_type carbon_footprint
## 3043                   Yes  Omnivore          10814.4
## 3093                    No     Vegan           7170.5
## 6043                    No  Omnivore          13117.7
## 6093                    No  Omnivore           9862.6
##      environmental_concern_score   income number_adults home_type
## 3043                           8       NA             1      Flat
## 3093                           4       NA             1  Detached
## 6043                           4 22353.31             6  Detached
## 6093                           3 12987.12             2      Flat
##      recycles_regularly_question
## 3043                         Yes
## 3093                          No
## 6043                         Yes
## 6093                          No
```

#Issue 2 : Missing Values in income

```
mydf[is.na(mydf$income), ]
```

```
##      household_size annual_energy_consumption recycles_regularly
## 2043              3                    6064.4                  1
## 2093              3                    3865.5                  0
## 2143              1                    3019.1                  1
## 2193              3                    5476.6                  0
## 2243              4                    4088.9                  0
## 2293              5                    4405.7                  0
## 2343              3                    4090.0                  1
## 2393              3                    5125.1                  0
## 2443              3                    5060.3                  0
## 2493              4                    4890.5                  1
## 2543              1                    3673.4                  0
## 2593              3                    5532.5                  1
## 2643              3                    5269.7                  0
## 2693              1                    4865.0                  0
## 2743              2                    3967.2                  1
## 2793              2                    4716.7                  1
## 2843              3                    4443.7                  1
## 2893              2                    4856.1                  0
## 2943              1                    3904.5                  1
## 2993              2                    2268.9                  1
## 3043              1                    -500.0                  1
## 3093              1                    -500.0                  0
##      uses_public_transport   diet_type carbon_footprint
## 2043                   Yes    Omnivore          13699.0
## 2093                    No Pescatarian          11062.6
## 2143                    No    Omnivore           7630.6
## 2193                    No  Vegetarian          15000.0
## 2243                   Yes    Omnivore           9862.2
## 2293                    No    Omnivore          14325.1
## 2343                   Yes    Omnivore           9232.5
## 2393                    No    Omnivore          12288.1
## 2443                    No    Omnivore          14900.5
```


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```
## 2493                   Yes Pescatarian          11658.7
## 2543                    No    Omnivore           9659.8
## 2593                   Yes    Omnivore          15000.0
## 2643                    No    Omnivore          14561.6
## 2693                    No Pescatarian          11958.5
## 2743                   Yes  Vegetarian           9346.1
## 2793                    No  Vegetarian          10951.6
## 2843                    No    Omnivore          10899.8
## 2893                    No       Vegan           9895.2
## 2943                   Yes    Omnivore           9185.3
## 2993                   Yes    Omnivore           5892.4
## 3043                   Yes    Omnivore          10814.4
## 3093                    No       Vegan           7170.5
##      environmental_concern_score income number_adults     home_type
## 2043                           9     NA              3      Terrrace
## 2093                           4     NA              3      Terrrace
## 2143                           6     NA              1 Semi-detached
## 2193                           4     NA              3      Detached
## 2243                           6     NA              4          Flat
## 2293                           5     NA              5          Flat
## 2343                           7     NA              3          Flat
## 2393                           5     NA              3      Terraced
## 2443                           3     NA              3      Terraced
## 2493                           7     NA              4          Flat
## 2543                           2     NA              0      Detached
## 2593                           6     NA              3      Terraced
## 2643                           4     NA              3      Terraced
## 2693                           4     NA              1      Terraced
## 2743                           7     NA              2      Terraced
## 2793                           5     NA              2          Flat
## 2843                           5     NA              3          Flat
## 2893                           6     NA              2      Detached
## 2943                           7     NA              1      Terraced
## 2993                           6     NA              2      Terraced
## 3043                           8     NA              1          Flat
## 3093                           4     NA              1      Detached
##      recycles_regularly_question
## 2043                         Yes
## 2093                          No
## 2143                         Yes
## 2193                          No
## 2243                          No
## 2293                          No
## 2343                         Yes
## 2393                          No
## 2443                          No
## 2493                         Yes
## 2543                          No
## 2593                         Yes
## 2643                          No
## 2693                          No
## 2743                         Yes
## 2793                         Yes
## 2843                         Yes
```


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```
## 2893                          No
## 2943                         Yes
## 2993                         Yes
## 3043                         Yes
## 3093                          No
```

```
sum(is.na(mydf$income))
```

```
## [1] 22
```

#Issue 3 : Misspellings and inconsistent labels in home_type

```
unique(mydf$home_type)
```

```
## [1] "Detached"      "Semi-detached" "Terraced"      "Flat"
## [5] "flat"          "Terrrace"      "semi-detached" "detatched"
```

```
table(mydf$home_type)
```

```
##
##      Detached     detatched          flat          Flat semi-detached
##            58             2             2            59             2
## Semi-detached      Terraced      Terrrace
##            52            63             2
```

#Issue 4 - Inconsistency between household_size and number_adults

```
sum(mydf$number_adults == 0)
```

```
## [1] 28
```

```
testmydf <- mydf[mydf$number_adults != 0, ]
all(testmydf$household_size == testmydf$number_adults)
```

```
## [1] TRUE
```

I identified this as a quality concern because some rows have 'number_adults' set to 0, while 'household_size' is 1, suggesting a household with only one minor and no adults, which is unusual given the income and other data. After removing the rows where 'number_adults' is 0, the remaining values in each column are consistent, suggesting a data entry error or something else.

1.4 Data cleaning

#Fix 1 : Implausible values in annual_energy_consumption

```
mydf <- mydf[mydf$annual_energy_consumption >= 1000 & mydf$annual_energy_consumption <= 9000, ]
```

#Fix 2 : fixing missing values

```r
median_income <- median(mydf$income, na.rm = TRUE)
mydf$income[is.na(mydf$income)] <- median_income
sum(is.na(mydf$income))
```

```
## [1] 0
```

#Fix 3 : Spelling error

```r
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "flat", "Flat")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "semi-detached", "Semi-detached")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "detatched", "Detached")
mydf$home_type <- replace(mydf$home_type, mydf$home_type == "Terrrace", "Terraced")

table(mydf$home_type)
```

```
##
##      Detached          Flat Semi-detached      Terraced
##            58            59            54            65
```

```r
unique(mydf$home_type)
```

```
## [1] "Detached"      "Semi-detached" "Terraced"      "Flat"
```

#Fix 4

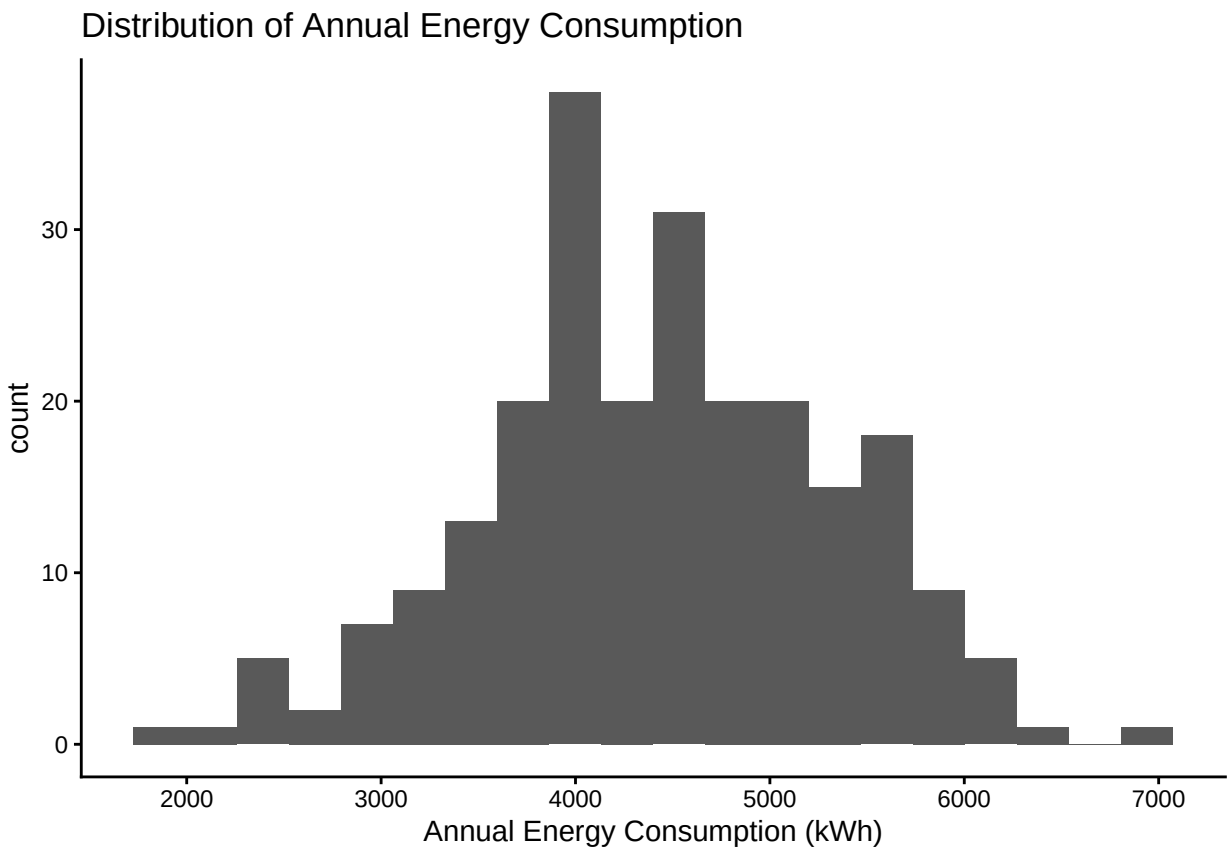
```r
mydf$number_adults[mydf$household_size == 1 & mydf$number_adults == 0] <- 1
sum(mydf$household_size == 1 & mydf$number_adults == 0)
```

```
## [1] 0
```

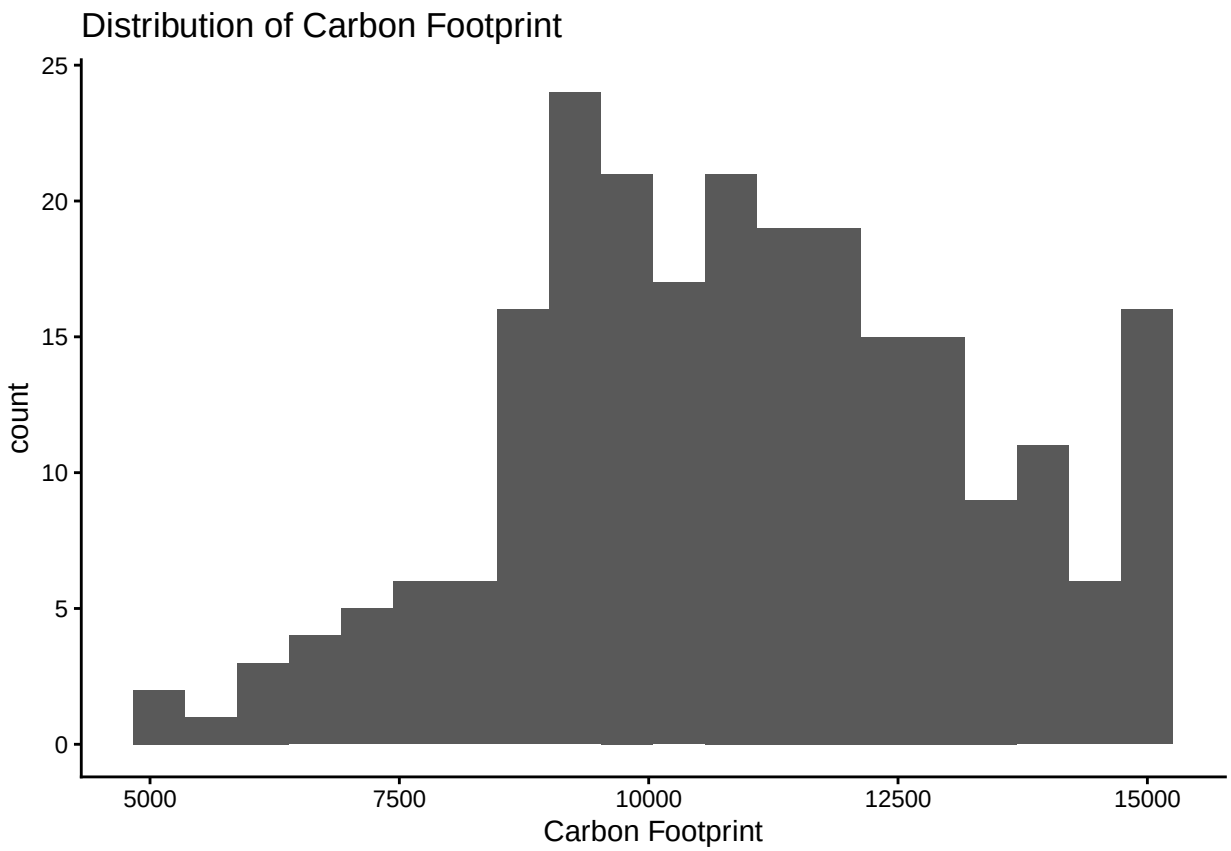
2. Exploratory Data Analysis (EDA)

#Elemenet 1 : Distribution

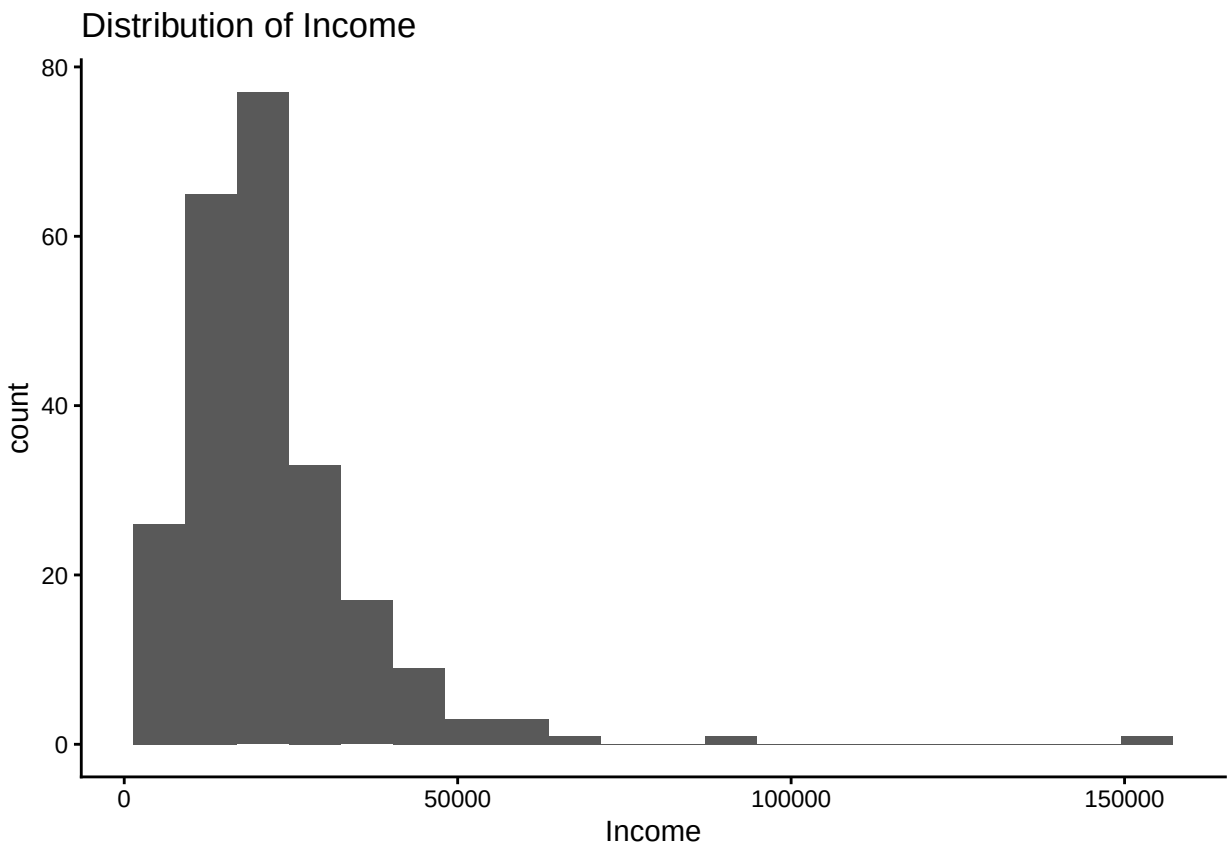
```r
ggplot(mydf, aes(x = annual_energy_consumption)) + geom_histogram(bins = 20) +ggtitle("Distribution of
```



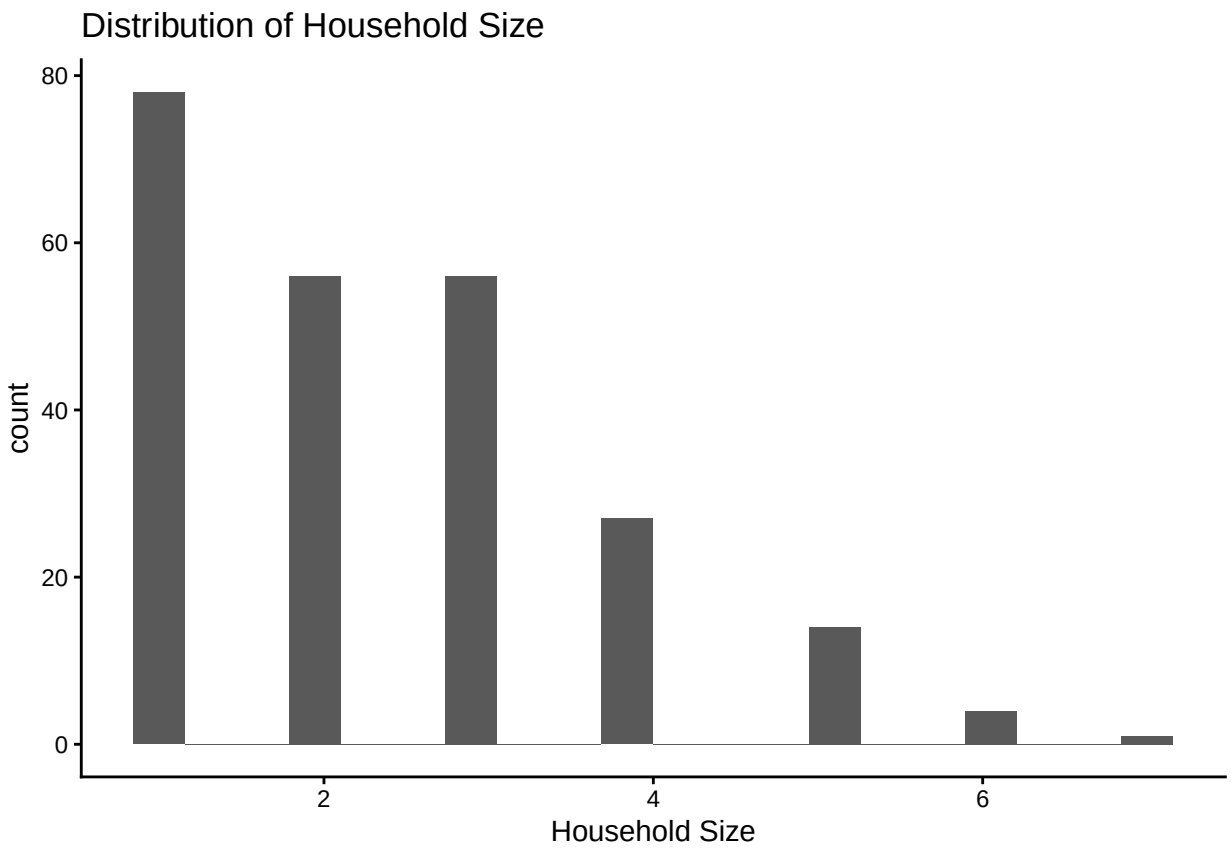

```
ggplot(mydf, aes(x = carbon_footprint)) + geom_histogram(bins = 20) + ggtitle("Distribution of Carbon Fo
xlab("Carbon Footprint") + theme_classic()
```



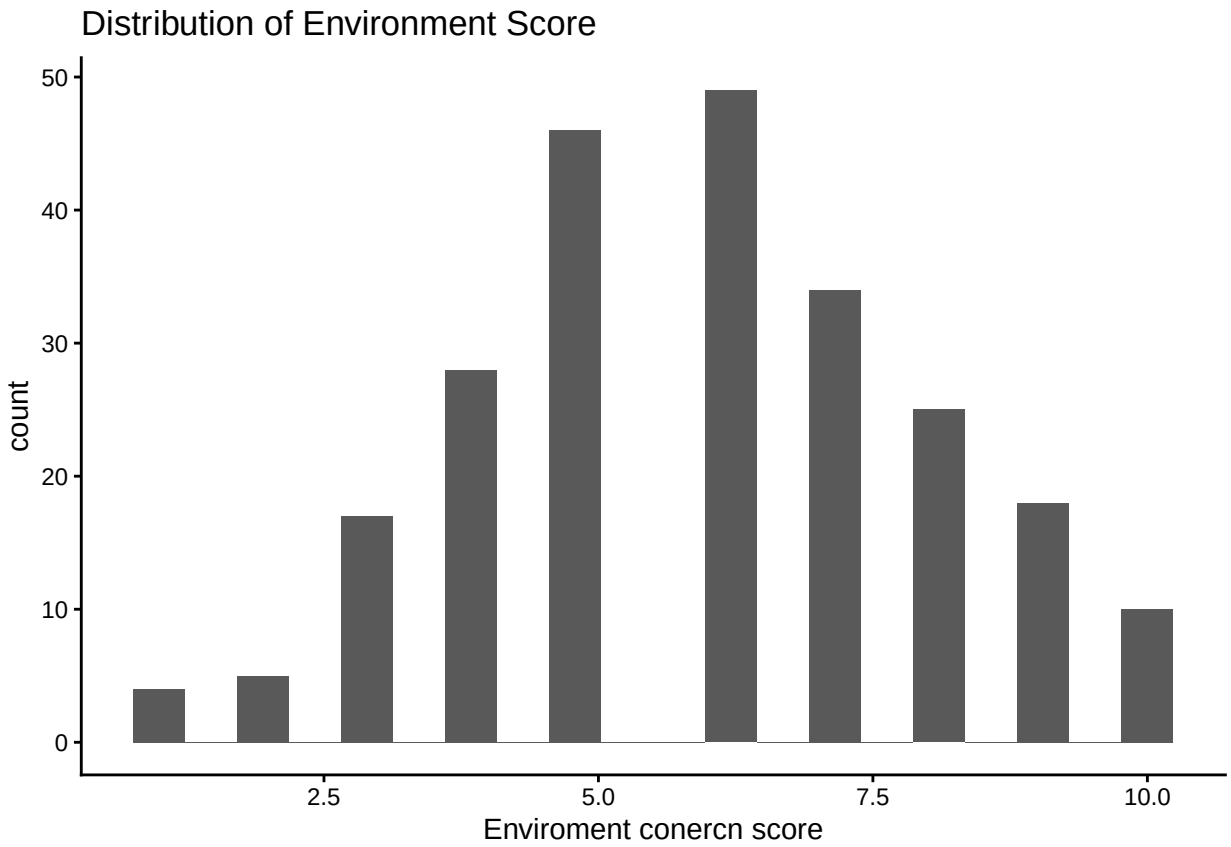

```
ggplot(mydf, aes(x = income)) + geom_histogram(bins = 20) + ggtitle("Distribution of Income") +
xlab("Income") + theme_classic()
```




```
ggplot(mydf, aes(x = household_size)) + geom_histogram(bins = 20) + ggtitle("Distribution of Household S
xlab("Household Size") + theme_classic()
```

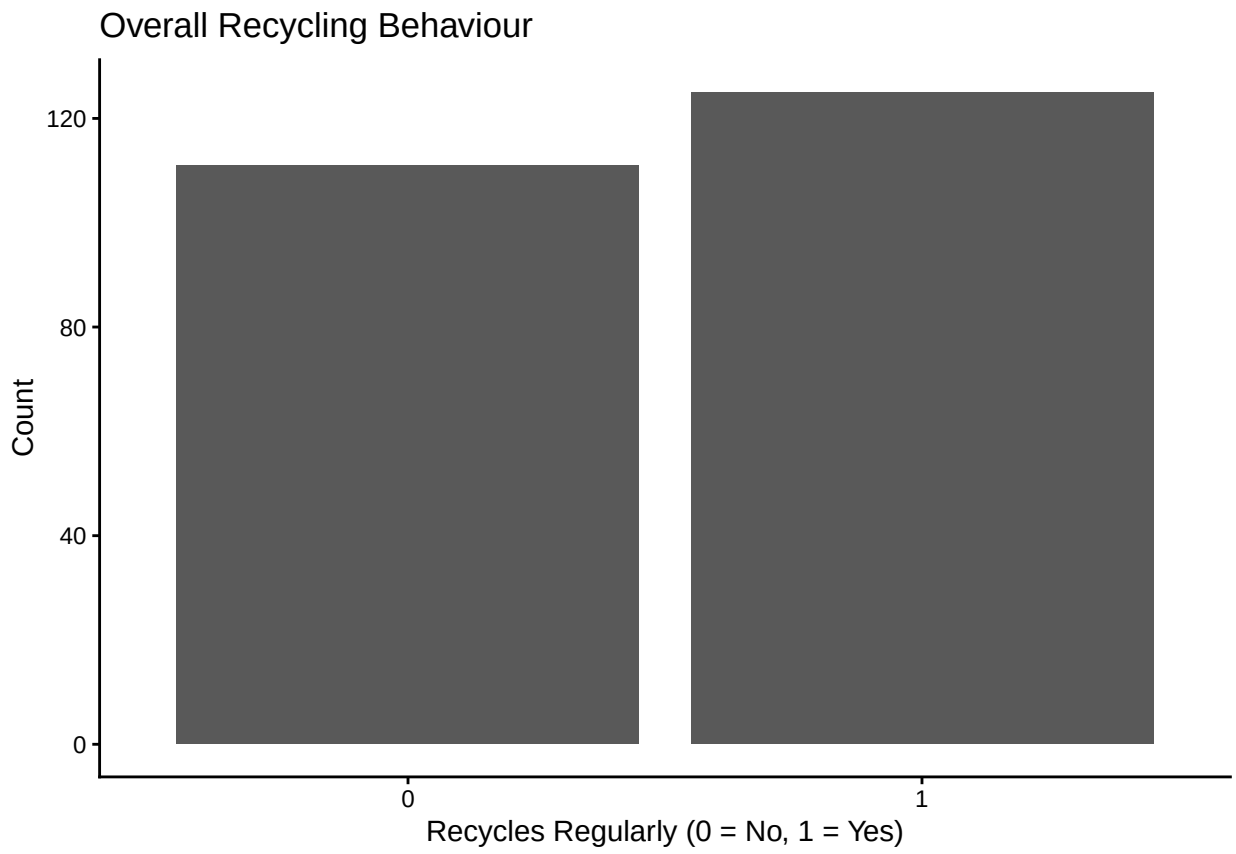



```
ggplot(mydf, aes(x = environmental_concern_score)) + geom_histogram(bins = 20) + ggtitle("Distribution
xlab("Enviroment conercn score") + theme_classic()
```

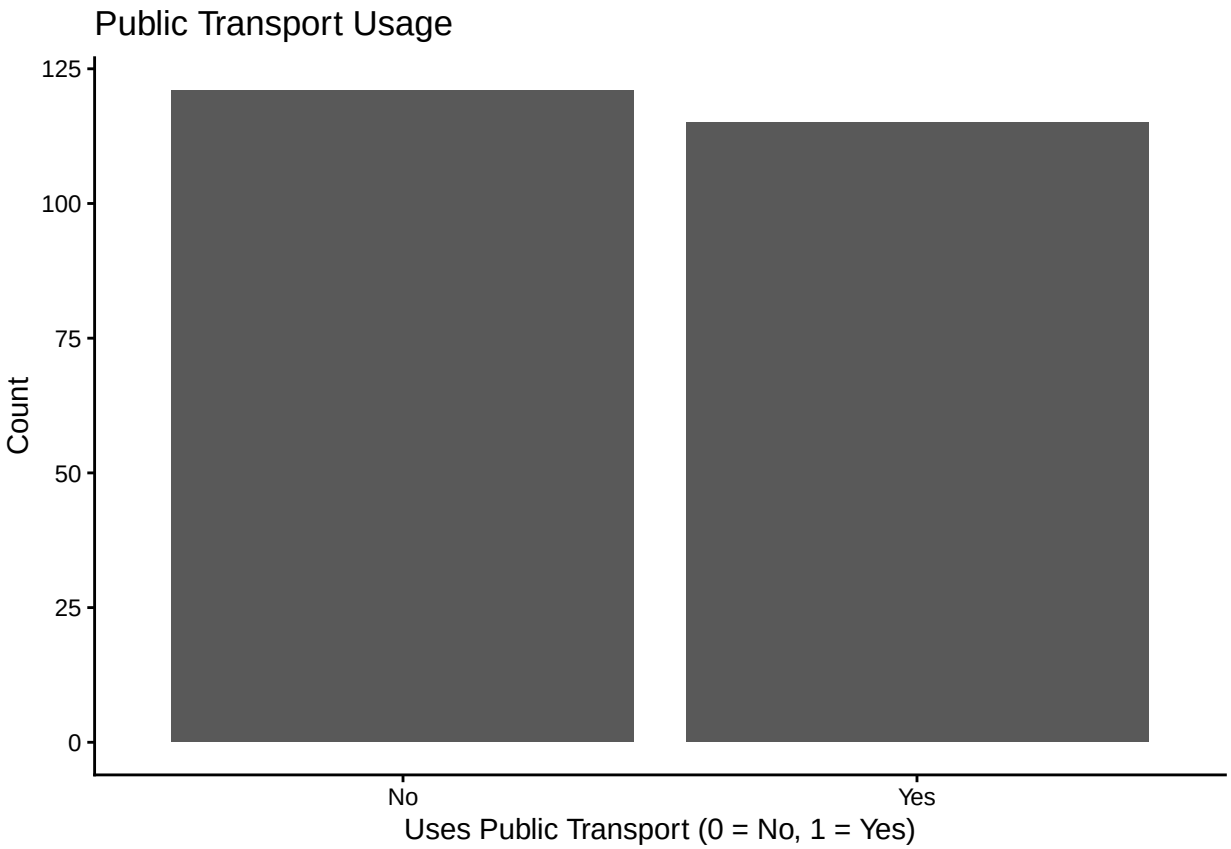


#Element 2 : Barplots

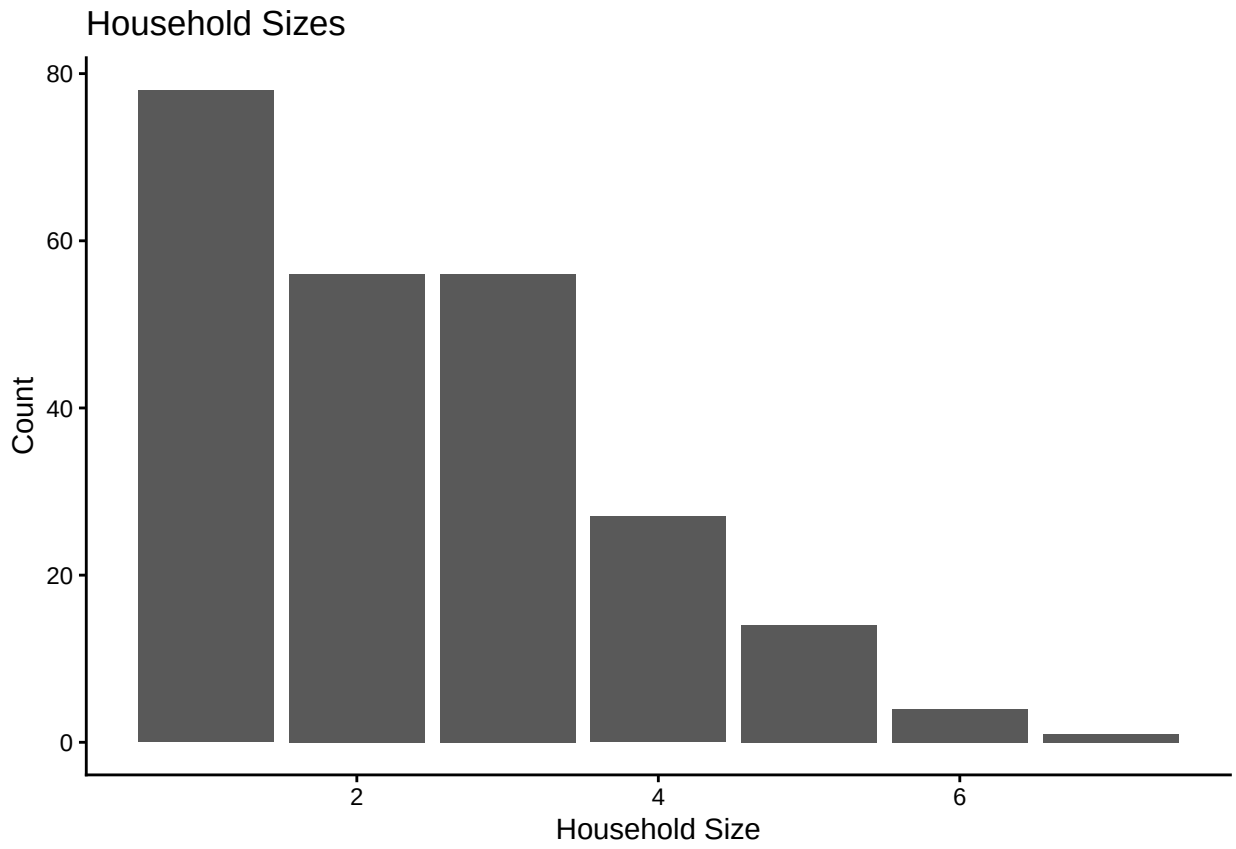
```
ggplot(mydf, aes(x = as.factor(recycles_regularly))) + geom_bar() + xlab("Recycles Regularly (0 = No, 1
```



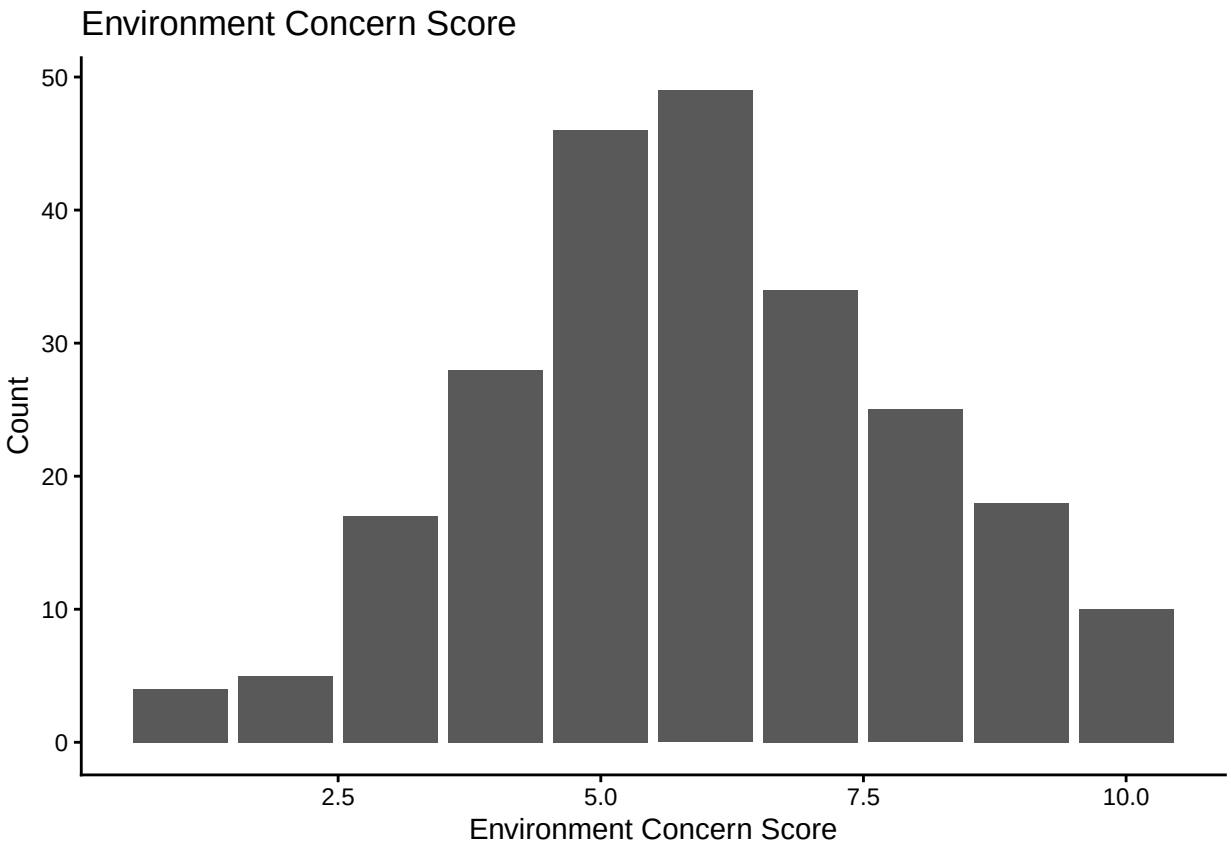

```
ggplot(mydf, aes(x = as.factor(uses_public_transport))) + geom_bar() + xlab("Uses Public Transport (0 =
```




```
ggplot(mydf, aes(x = household_size)) + geom_bar() + xlab("Household Size") + ylab("Count") + ggtitle("
```

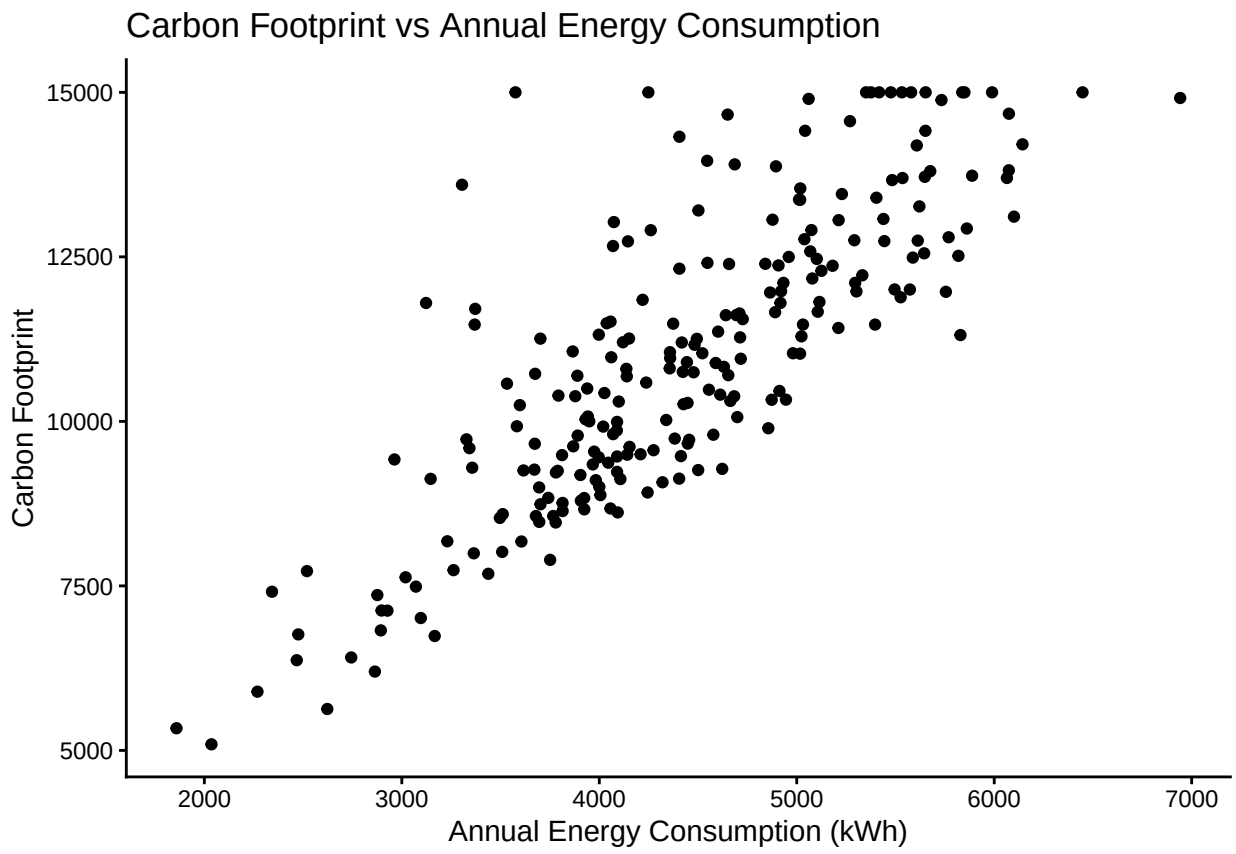



```
ggplot(mydf, aes(x = environmental_concern_score)) + geom_bar() + xlab("Environment Concern Score") + yl
```

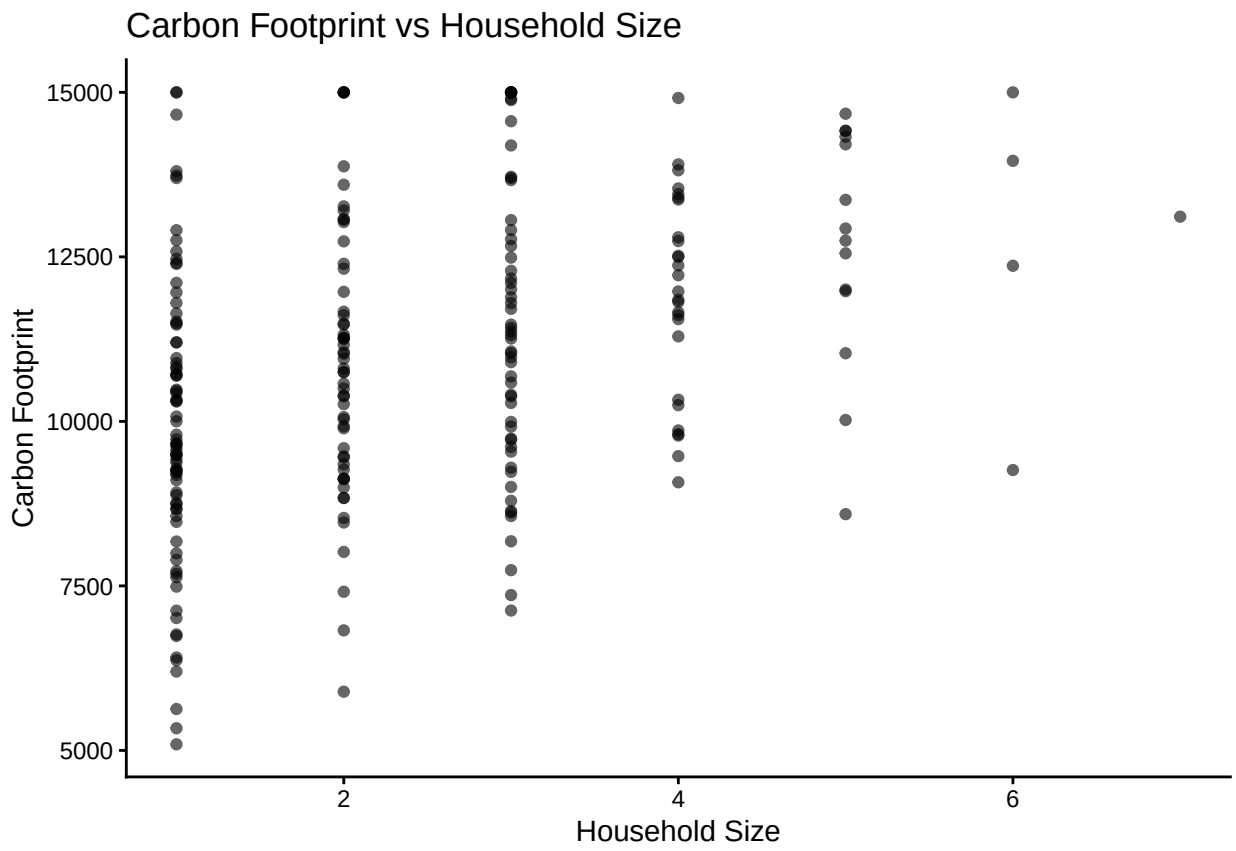


#Element 3 : Scatterplots

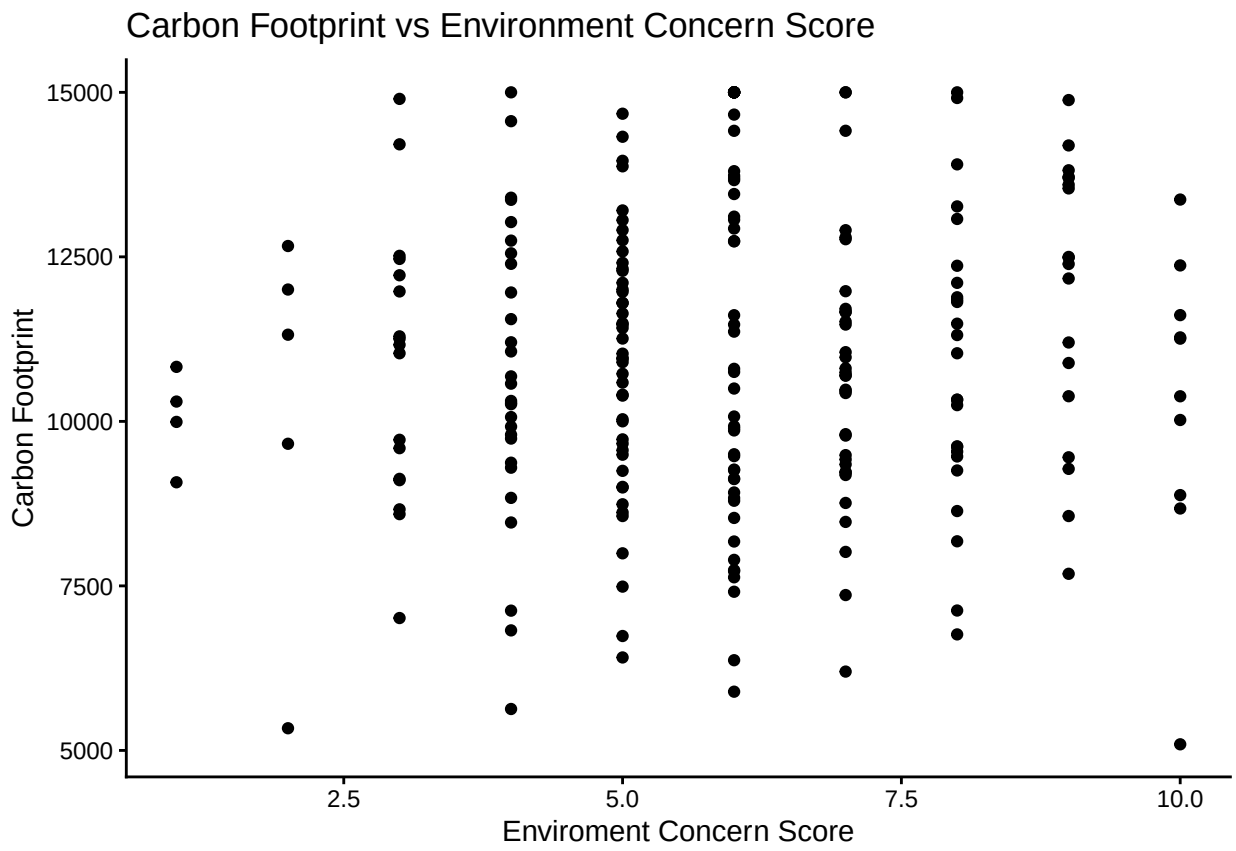
```
ggplot(mydf, aes(x = annual_energy_consumption, y = carbon_footprint)) + geom_point() + ggtitle("Carbon
```



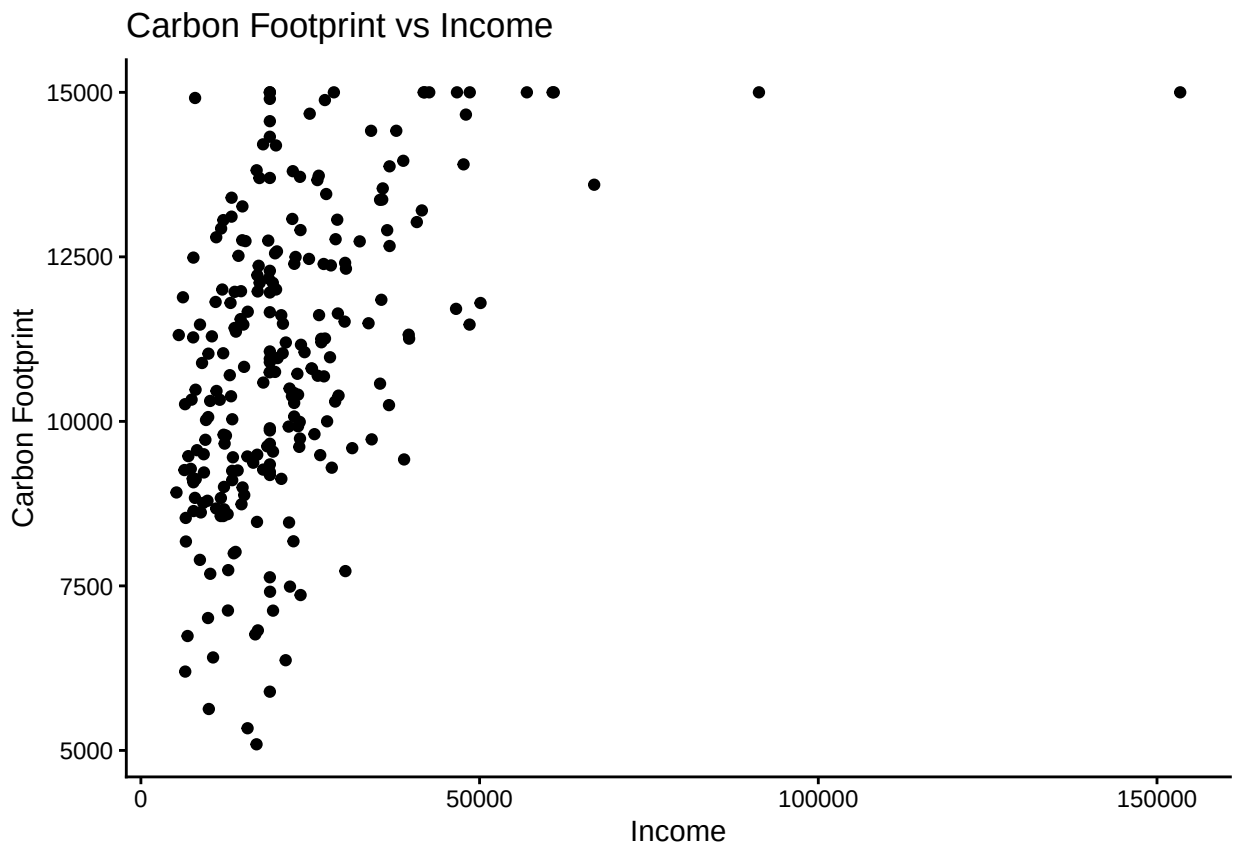

```
ggplot(mydf, aes(x = household_size, y = carbon_footprint)) + geom_point(alpha = 0.6) + ggtitle("Carbon
```




```
ggplot(mydf, aes(x = environmental_concern_score, y = carbon_footprint)) + geom_point() + ggtitle("Carb
```

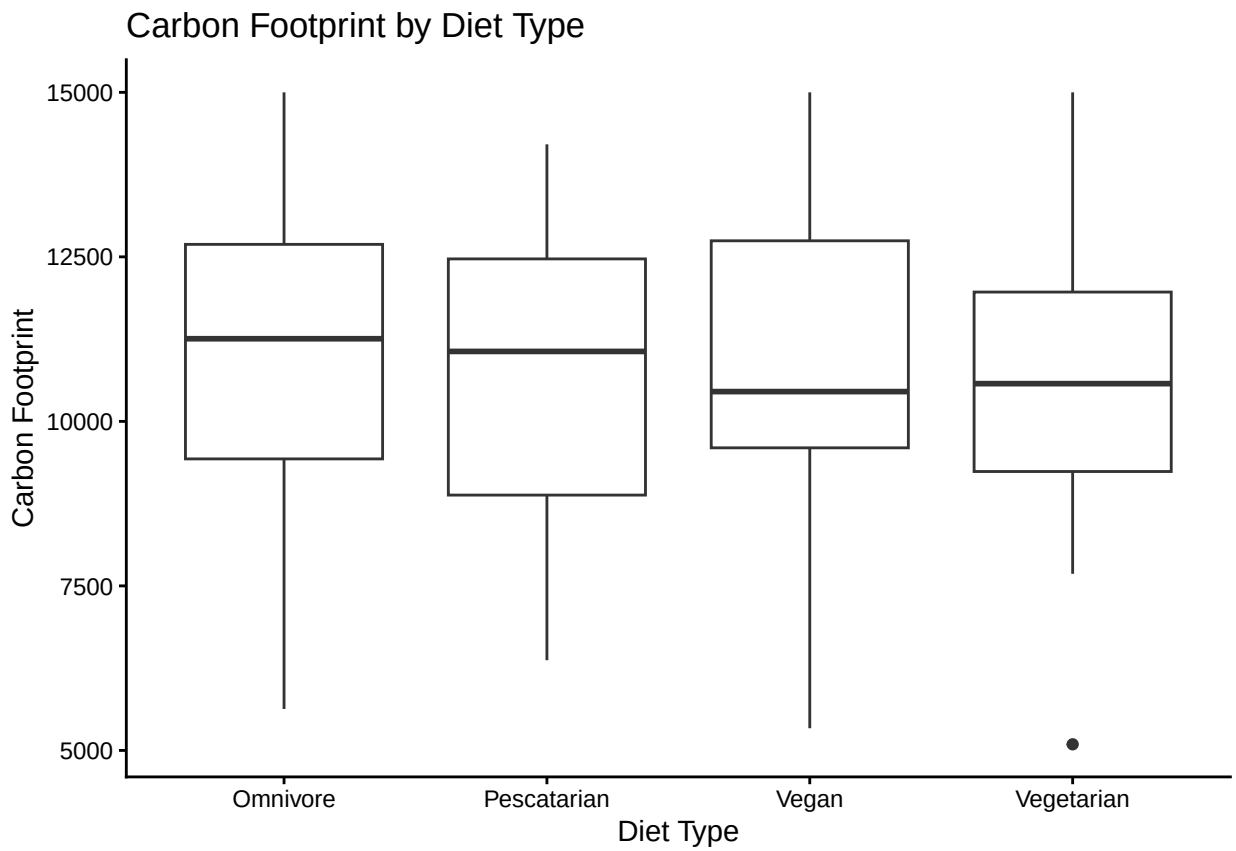



```
ggplot(mydf, aes(x = income, y = carbon_footprint)) + geom_point() + ggtitle("Carbon Footprint vs Income
```

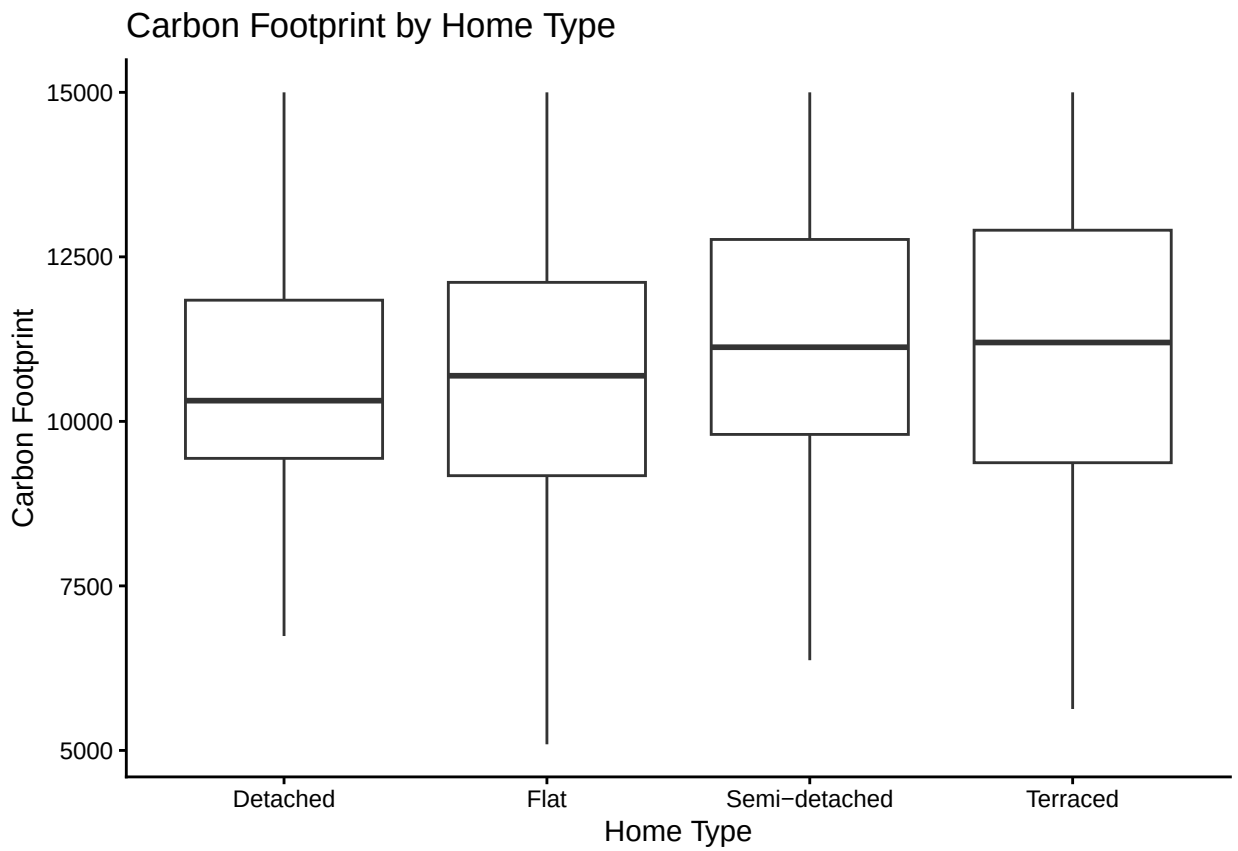


#Element 4 : Boxplots

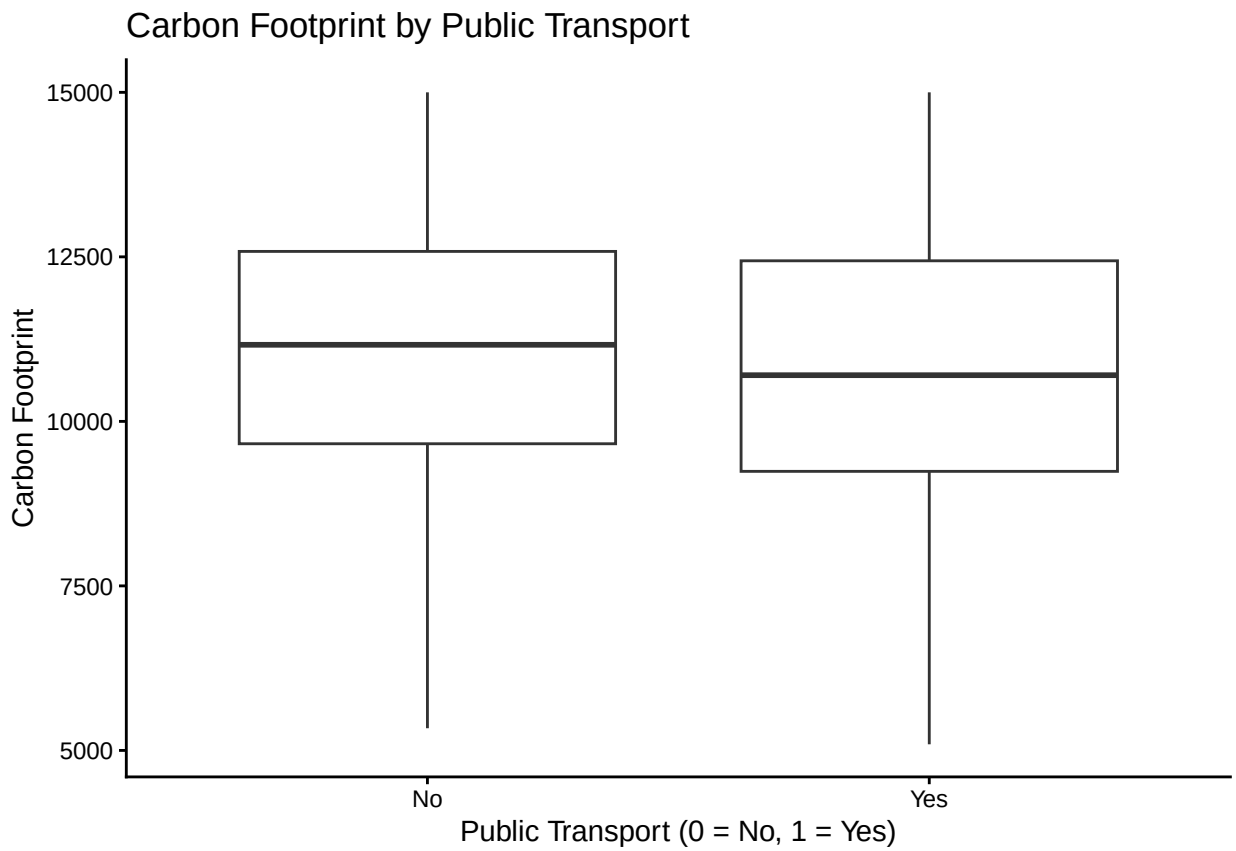
```
ggplot(mydf, aes(x = diet_type, y = carbon_footprint)) + geom_boxplot() + ggtitle("Carbon Footprint by
```



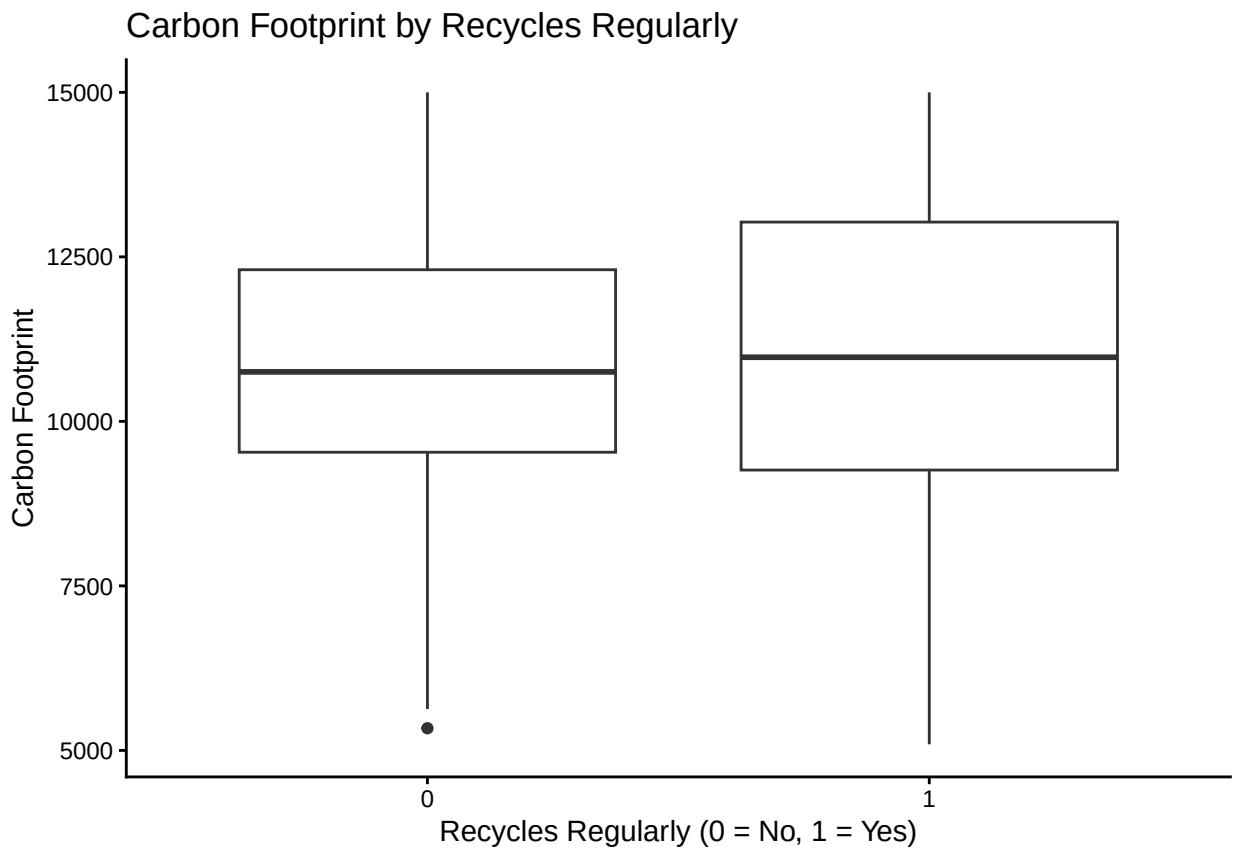

```
ggplot(mydf, aes(x = home_type, y = carbon_footprint)) + geom_boxplot() + ggtitle("Carbon Footprint by H
xlab("Home Type") + ylab("Carbon Footprint") + theme_classic()
```



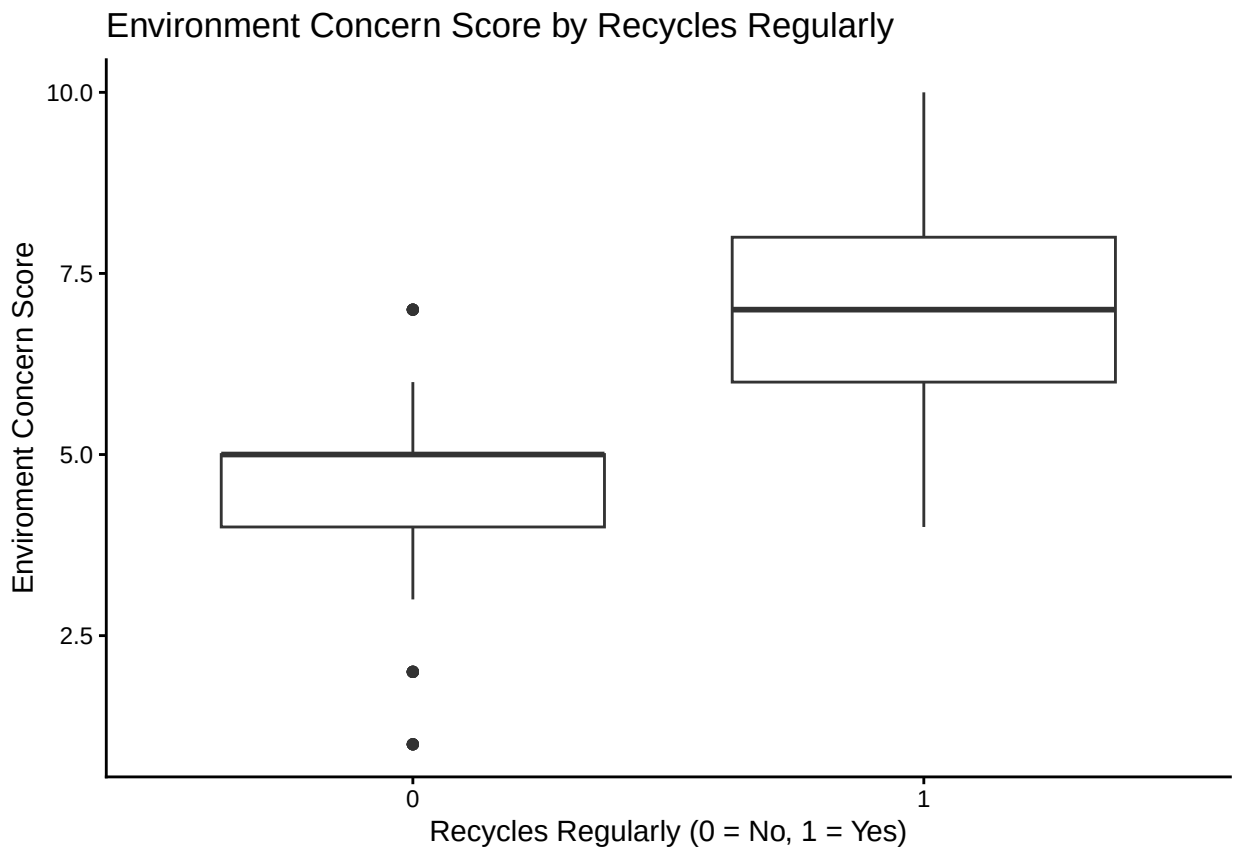

```
ggplot(mydf, aes(x = as.factor(uses_public_transport), y = carbon_footprint)) + geom_boxplot() + ggtitle
```



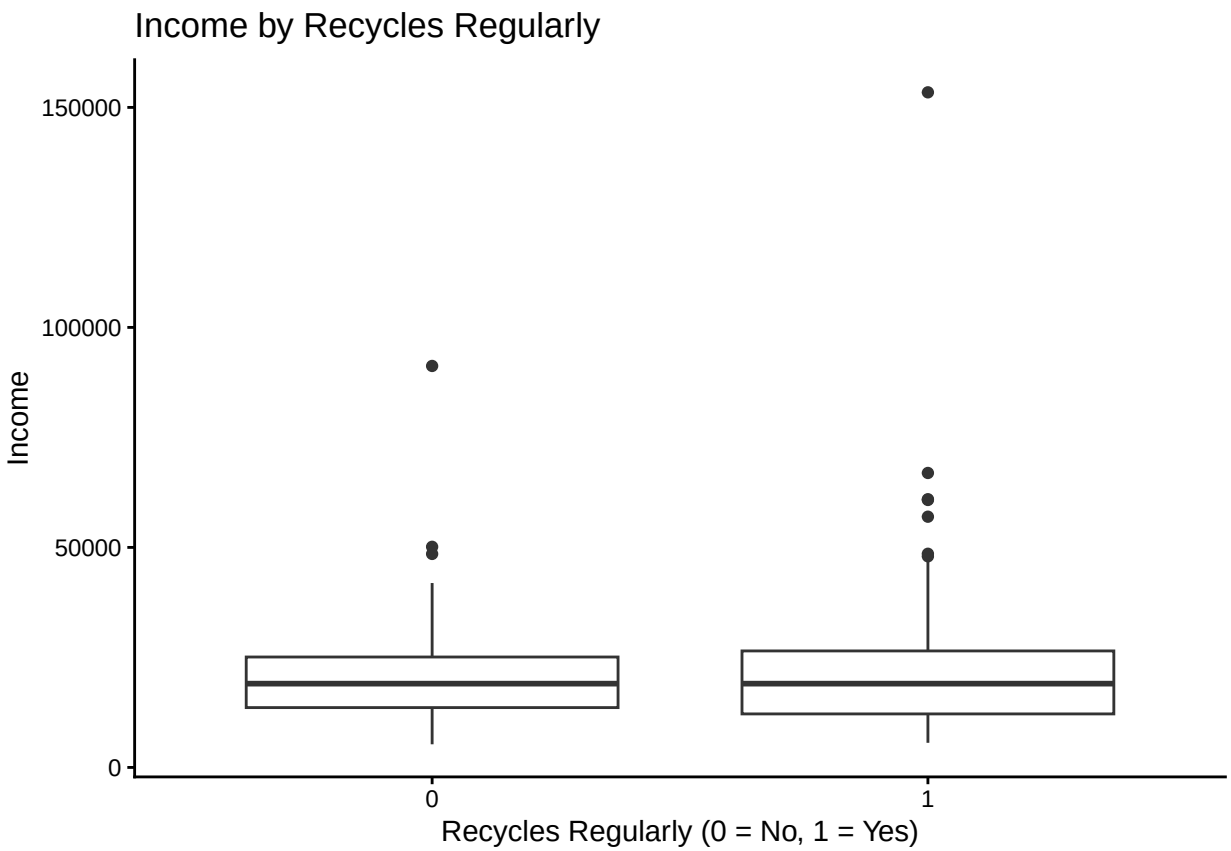

```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = carbon_footprint)) + geom_boxplot() + ggtitle("
```



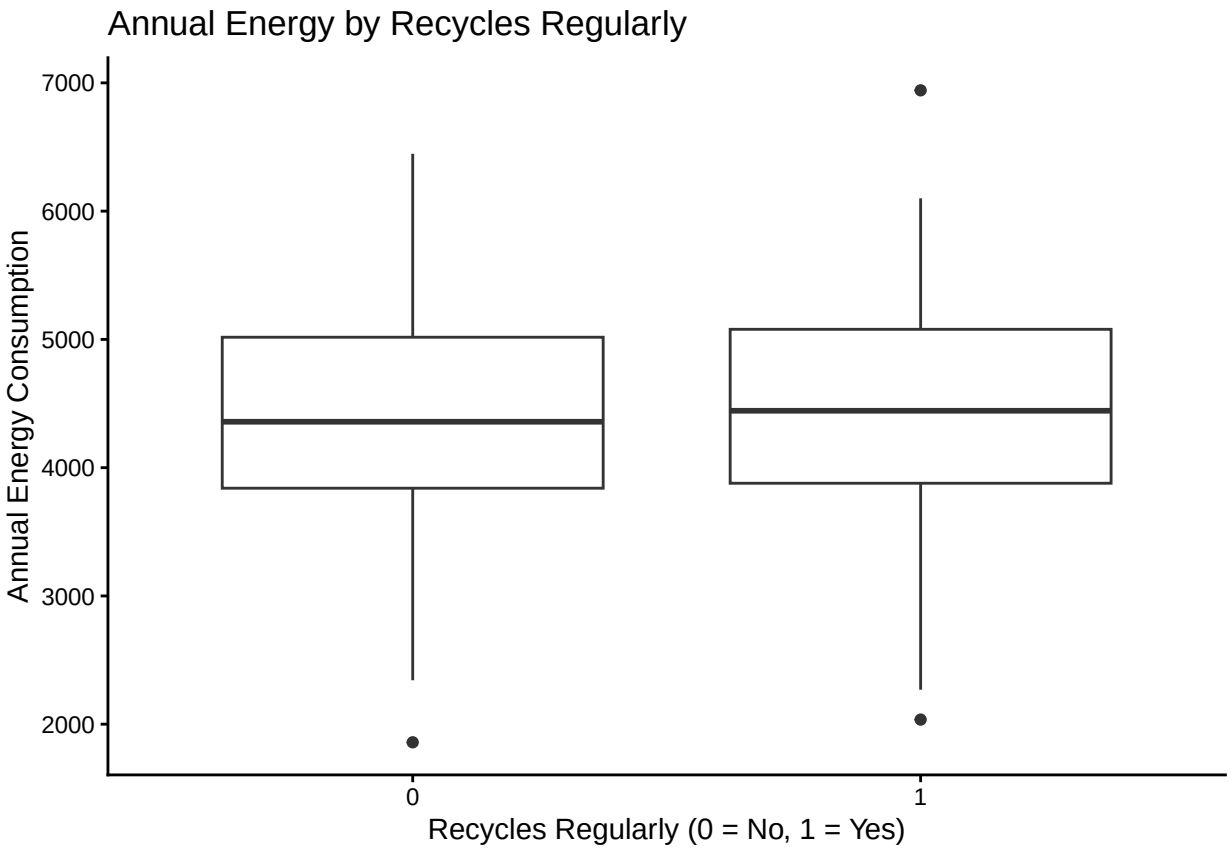

```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = environmental_concern_score)) + geom_boxplot() +
```



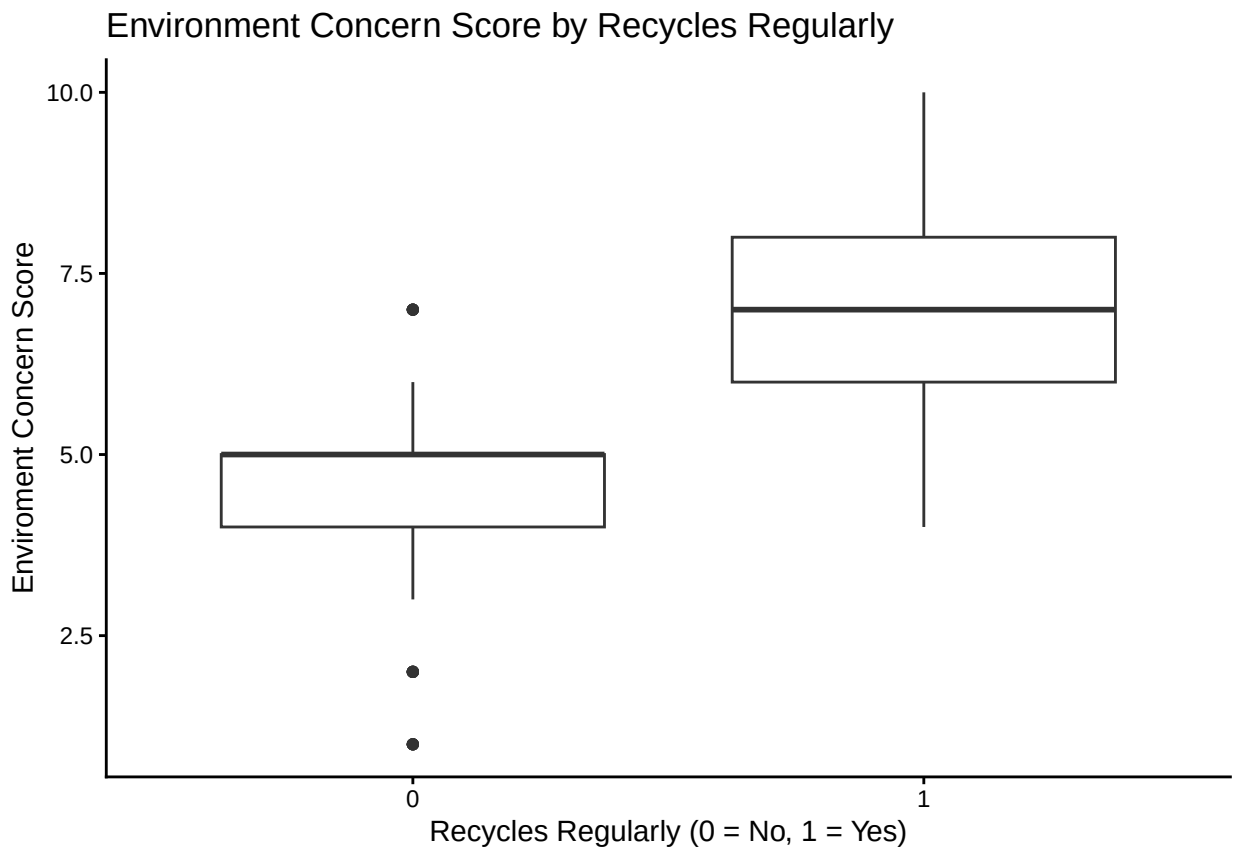

```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = income)) + geom_boxplot() + ggtitle("Income by
```



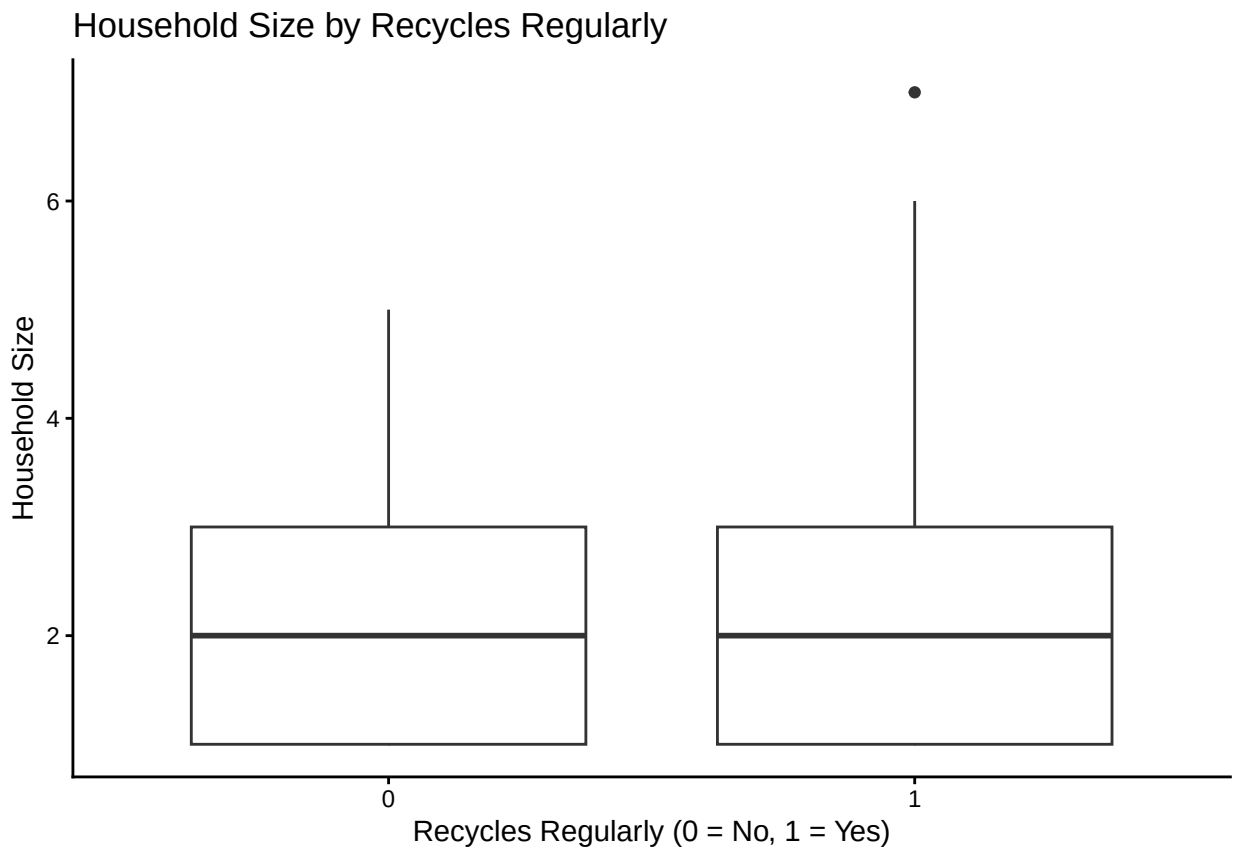

```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = annual_energy_consumption)) + geom_boxplot() + g
```




```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = environmental_concern_score)) + geom_boxplot() +
```

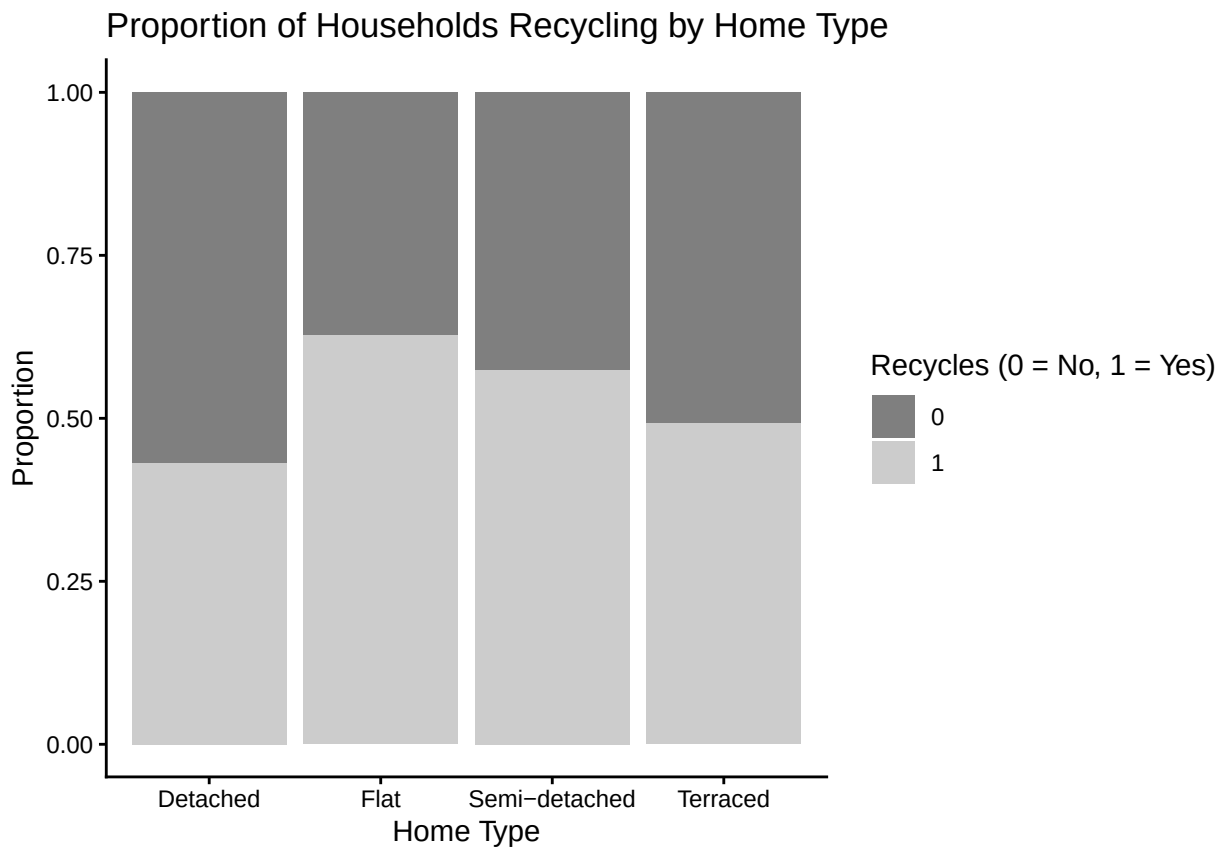



```
ggplot(mydf, aes(x = as.factor(recycles_regularly), y = household_size)) + geom_boxplot() + ggtitle("Ho
```

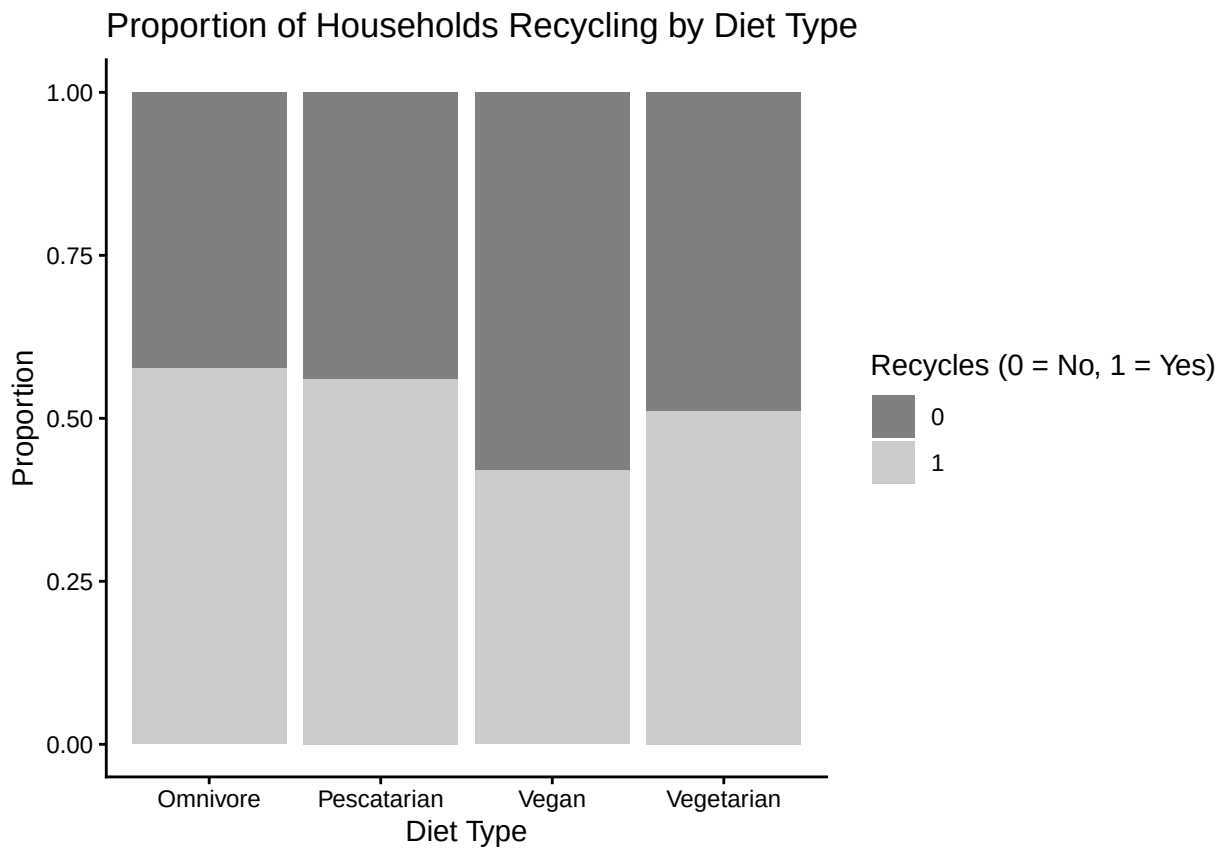


#Element 5 : Barplots Compare

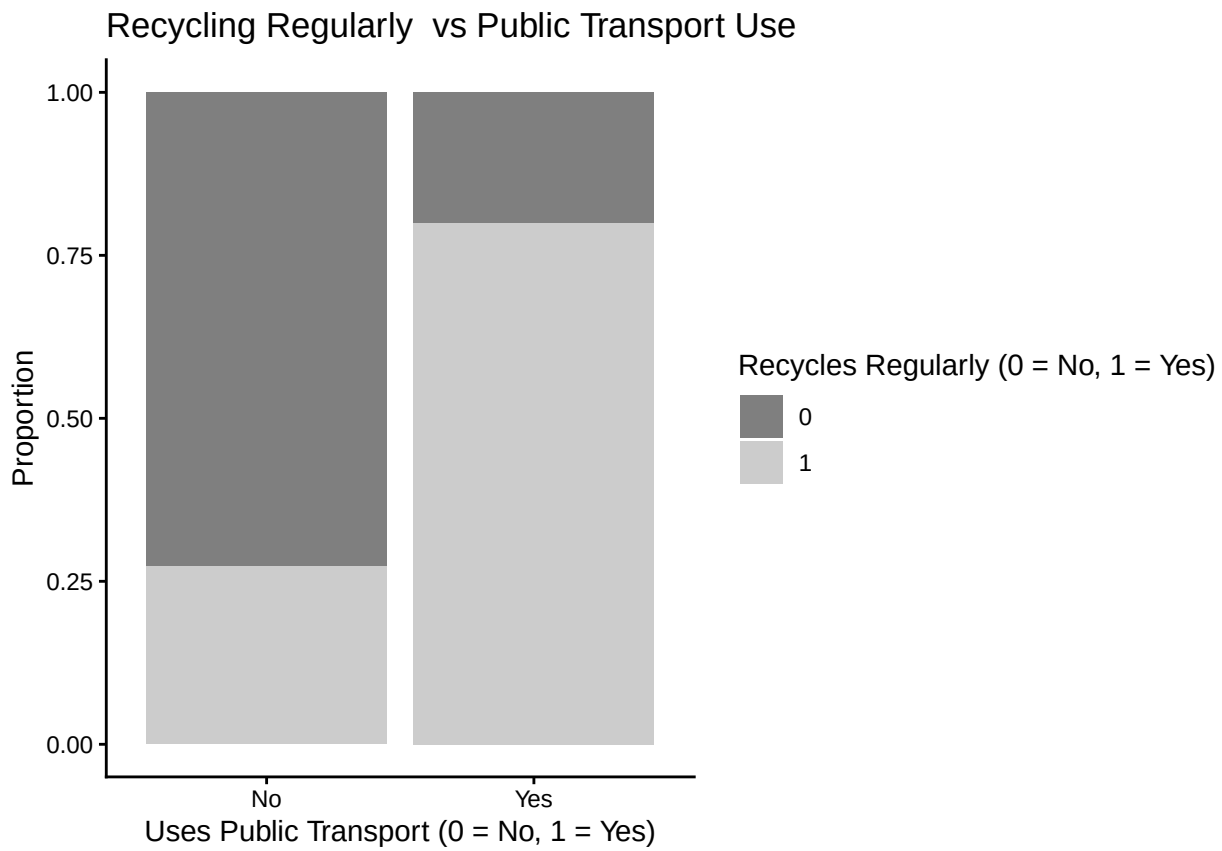
```
ggplot(mydf, aes(x = home_type, fill = as.factor(recycles_regularly))) + geom_bar(position = "fill") + 
```




```
ggplot(mydf, aes(x = diet_type, fill = as.factor(recycles_regularly))) + geom_bar(position = "fill") + 
```




```
ggplot(mydf, aes(x = as.factor(uses_public_transport), fill = as.factor(recycles_regularly))) + geom_ba
```



#Element 6 : Correlation

```
cor(mydf$annual_energy_consumption, mydf$carbon_footprint, use = "complete.obs")
```

```
## [1] 0.8179488
```

```
cor(mydf$income, mydf$carbon_footprint, use = "complete.obs")
```

```
## [1] 0.4682777
```

```
cor(mydf$household_size, mydf$carbon_footprint, use = "complete.obs")
```

```
## [1] 0.387023
```

```
cor(mydf$environmental_concern_score, mydf$carbon_footprint, use = "complete.obs")
```

```
## [1] 0.06414217
```

3. Modelling

#Initial Model

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```r
# Maximal model informed by EDA and metadata
m1.lm <- lm(carbon_footprint ~ annual_energy_consumption + income + household_size + environmental_conc
summary(m1.lm)
```

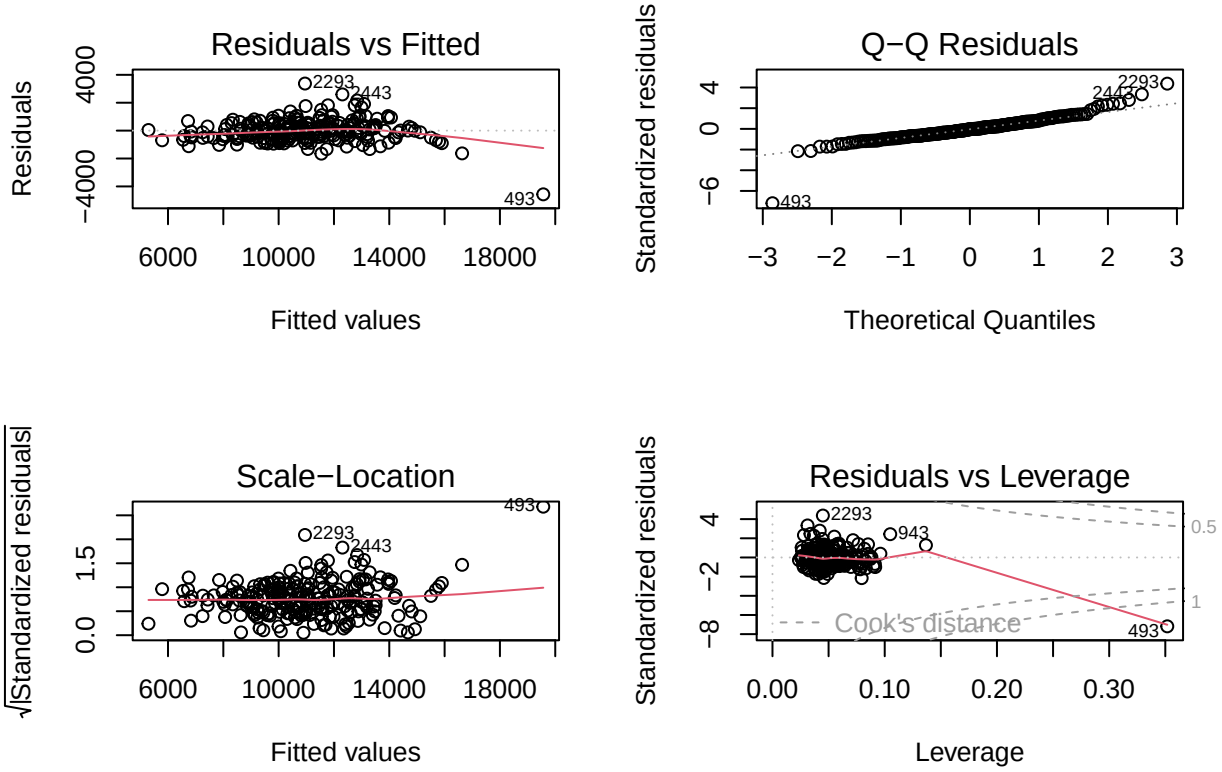
```
##
## Call:
## lm(formula = carbon_footprint ~ annual_energy_consumption + income +
##     household_size + environmental_concern_score + diet_type +
##     uses_public_transport + home_type, data = mydf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4567.3  -463.7   -26.3   408.1  3373.3
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  3.176e+02  3.305e+02   0.961    0.338
## annual_energy_consumption    1.984e+00  6.610e-02  30.022   <2e-16 ***
## income                       6.902e-02  3.519e-03  19.616   <2e-16 ***
## household_size               6.471e+01  4.403e+01   1.470    0.143
## environmental_concern_score  2.960e+01  3.744e+01   0.791    0.430
## diet_typePescatarian         2.938e+01  1.758e+02   0.167    0.867
## diet_typeVegan              -2.951e+01  1.350e+02  -0.218    0.827
## diet_typeVegetarian          2.821e+01  1.448e+02   0.195    0.846
## uses_public_transportYes    -2.432e+02  1.493e+02  -1.629    0.105
## home_typeFlat                1.059e+02  1.520e+02   0.697    0.487
## home_typeSemi-detached       1.384e+02  1.526e+02   0.907    0.365
## home_typeTerraced            3.491e+02  1.451e+02   2.405    0.017 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 789.5 on 224 degrees of freedom
## Multiple R-squared:  0.8828, Adjusted R-squared:  0.8771
## F-statistic: 153.4 on 11 and 224 DF,  p-value: < 2.2e-16
```

```r
exp(coef(m1.lm))
```

```
##                 (Intercept)   annual_energy_consumption
##               8.466688e+137                7.274570e+00
##                      income              household_size
##                1.071462e+00                1.269921e+28
## environmental_concern_score        diet_typePescatarian
##                7.178452e+12                5.748615e+12
##              diet_typeVegan         diet_typeVegetarian
##                1.532176e-13                1.790878e+12
##    uses_public_transportYes                home_typeFlat
##               2.395921e-106                9.484401e+45
##      home_typeSemi-detached           home_typeTerraced
##                1.332767e+60               3.921925e+151
```

###Initial Model Diagnistics

```r
par(mfrow=c(2,2))
plot(m1.lm)
```




```r
par(mfrow=c(1,1))
```

#Proposed Model

```r
m2.lm <- step(m1.lm)
```

```
## Start:  AIC=3160.61
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     environmental_concern_score + diet_type + uses_public_transport +
##     home_type
##
##                                Df Sum of Sq       RSS    AIC
## - diet_type                     3     93942 139732118 3154.8
## - environmental_concern_score   1    389704 140027880 3159.3
## <none>                                      139638176 3160.6
## - household_size                1   1346443 140984619 3160.9
## - home_type                     3   3878661 143516837 3161.1
## - uses_public_transport         1   1653630 141291806 3161.4
## - income                        1 239876295 379514471 3394.6
## - annual_energy_consumption     1 561858851 701497027 3539.6
##
```

```
## Step:  AIC=3154.77
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     environmental_concern_score + uses_public_transport + home_type
##
##                               Df Sum of Sq       RSS    AIC
## - environmental_concern_score  1    383332 140115450 3153.4
## <none>                                     139732118 3154.8
## - household_size               1   1501260 141233378 3155.3
## - home_type                    3   3963067 143695185 3155.4
## - uses_public_transport        1   1647791 141379910 3155.5
## - income                       1 240438459 380170577 3389.0
## - annual_energy_consumption    1 568995205 708727323 3536.0
##
## Step:  AIC=3153.42
## carbon_footprint ~ annual_energy_consumption + income + household_size +
##     uses_public_transport + home_type
##
##                             Df Sum of Sq       RSS    AIC
## <none>                                   140115450 3153.4
## - household_size             1   1433584 141549034 3153.8
## - home_type                  3   3858597 143974047 3153.8
## - uses_public_transport      1   1465020 141580470 3153.9
## - income                     1 243412043 383527493 3389.1
## - annual_energy_consumption  1 583120527 723235977 3538.8
```

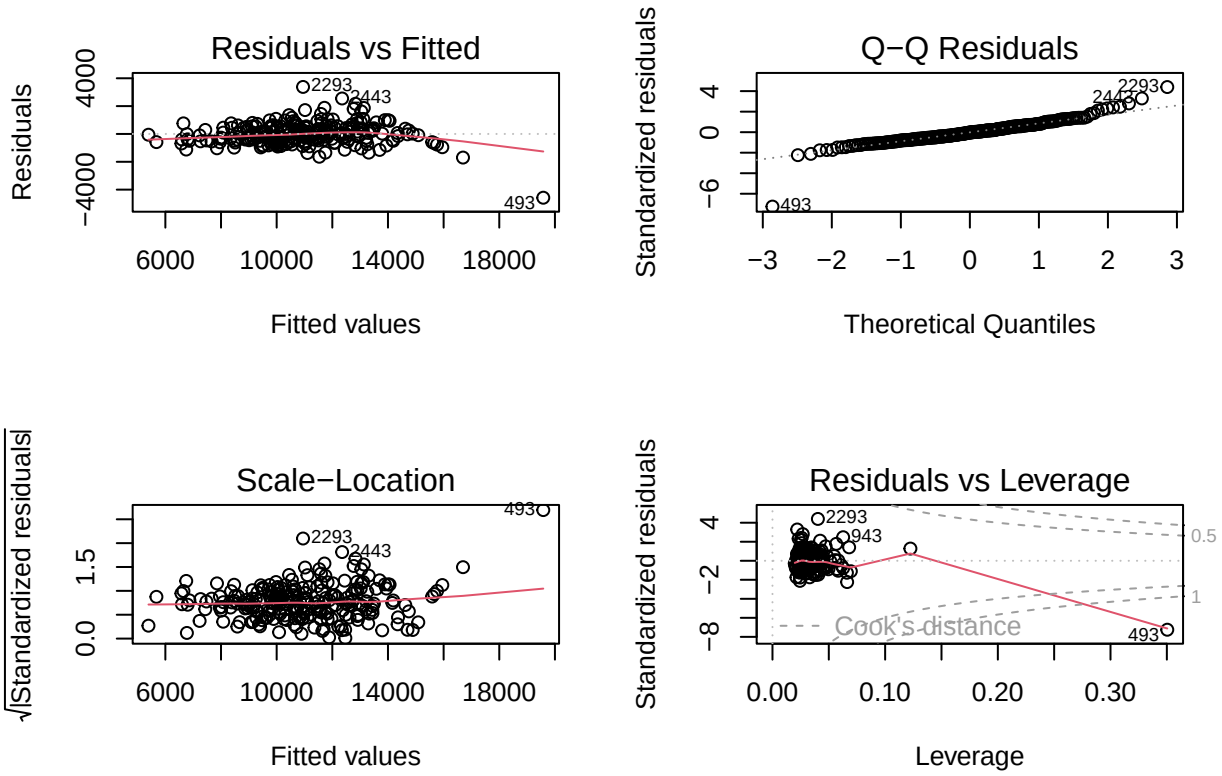
#Proposed Model Diagonistics

```
summary(m2.lm)
```

```
##
## Call:
## lm(formula = carbon_footprint ~ annual_energy_consumption + income +
##     household_size + uses_public_transport + home_type, data = mydf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4581.3  -465.0   -27.7   441.6  3377.2
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                4.269e+02  2.909e+02   1.468   0.1436
## annual_energy_consumption  1.988e+00  6.455e-02  30.804   <2e-16 ***
## income                     6.921e-02  3.478e-03  19.902   <2e-16 ***
## household_size             6.579e+01  4.308e+01   1.527   0.1281
## uses_public_transportYes  -1.580e+02  1.023e+02  -1.544   0.1240
## home_typeFlat              1.137e+02  1.472e+02   0.773   0.4405
## home_typeSemi-detached     1.414e+02  1.491e+02   0.948   0.3440
## home_typeTerraced          3.475e+02  1.425e+02   2.439   0.0155 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 783.9 on 228 degrees of freedom
## Multiple R-squared:  0.8824, Adjusted R-squared:  0.8788
## F-statistic: 244.5 on 7 and 228 DF,  p-value: < 2.2e-16
```


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```
par(mfrow=c(2,2))
plot(m2.lm)
```




```
par(mfrow=c(1,1))
```

#Proposed Model Interperations

4. Modelling another dependent variable

#Proposed Model # Maximal logistic model

```
m4.glm <- glm( recycles_regularly ~ environmental_concern_score + diet_type + home_type + uses_public_t
```

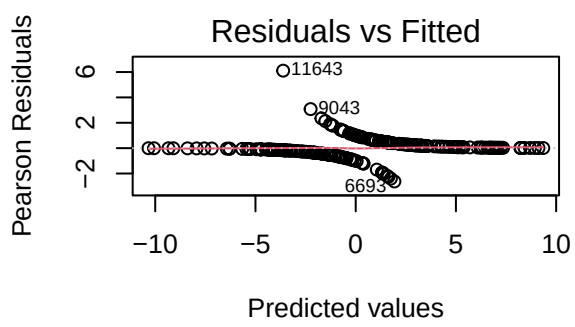
#Visuals

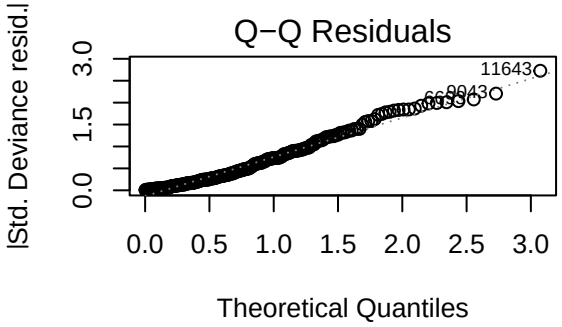
```
summary(m4.glm)
```

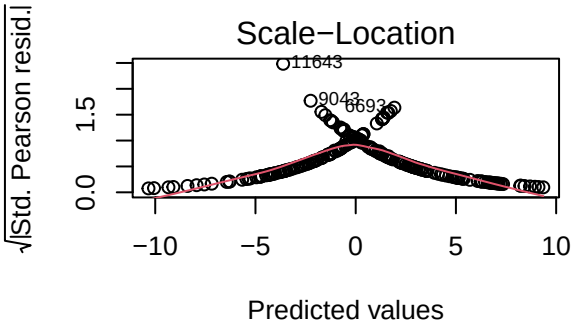
```
##
## Call:
## glm(formula = recycles_regularly ~ environmental_concern_score +
##     diet_type + home_type + uses_public_transport + annual_energy_consumption +
##     income + household_size, family = binomial, data = mydf)
```

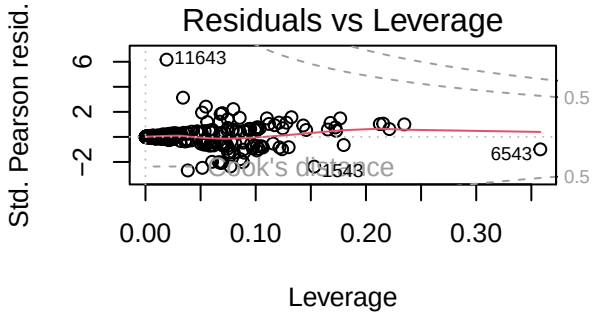
```
##
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 -1.281e+01  2.103e+00  -6.090 1.13e-09 ***
## environmental_concern_score  1.985e+00  3.132e-01   6.339 2.31e-10 ***
## diet_typePescatarian         1.847e-01  8.913e-01   0.207  0.83580
## diet_typeVegan              -1.166e+00  5.867e-01  -1.987  0.04688 *
## diet_typeVegetarian          4.862e-01  5.720e-01   0.850  0.39531
## home_typeFlat                1.750e+00  6.497e-01   2.694  0.00706 **
## home_typeSemi-detached       1.108e+00  6.434e-01   1.722  0.08507 .
## home_typeTerraced            1.213e+00  6.433e-01   1.886  0.05936 .
## uses_public_transportYes    -4.933e-01  5.480e-01  -0.900  0.36803
## annual_energy_consumption    7.164e-05  2.675e-04   0.268  0.78882
## income                       7.261e-07  1.551e-05   0.047  0.96267
## household_size               1.700e-01  1.736e-01   0.979  0.32749
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 326.33  on 235  degrees of freedom
## Residual deviance: 141.56  on 224  degrees of freedom
## AIC: 165.56
##
## Number of Fisher Scoring iterations: 6
```

```r
par(mfrow=c(2,2))
plot(m4.glm)
```










```
par(mfrow=c(1,1))
exp(coef(m4.glm))
```

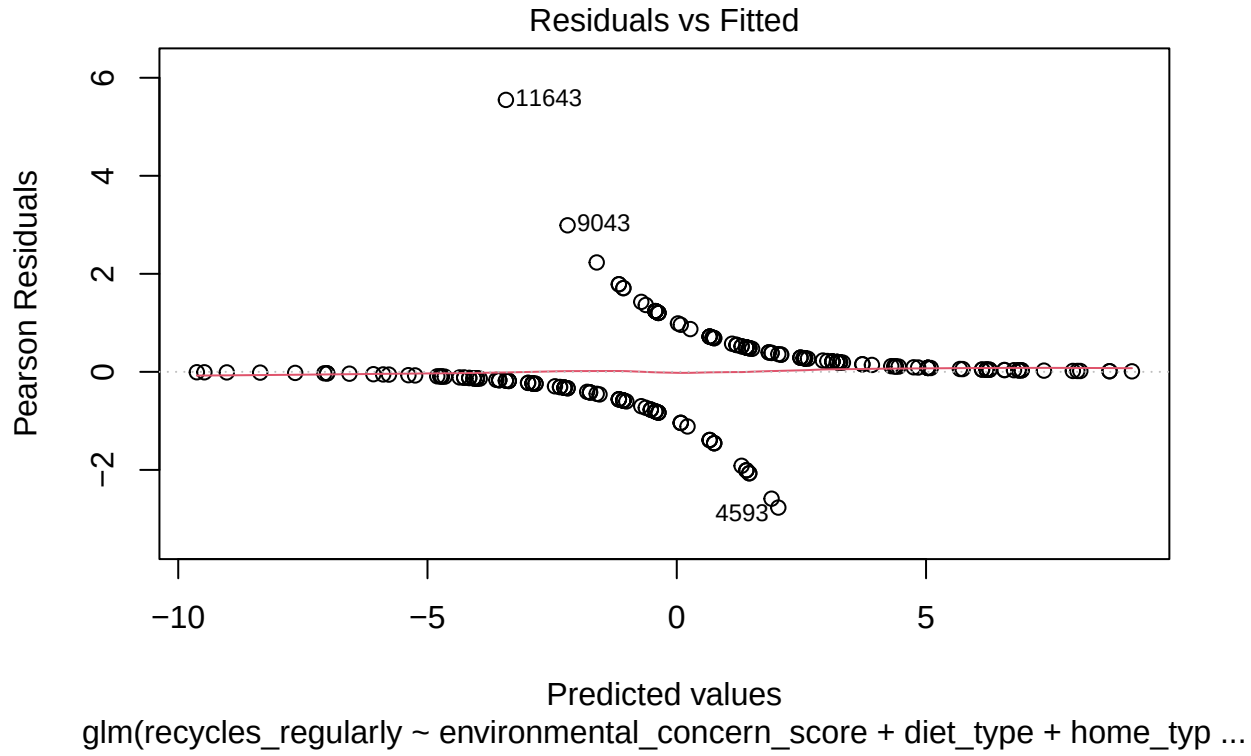
```
##                 (Intercept) environmental_concern_score
##                2.741756e-06                7.282636e+00
##        diet_typePescatarian              diet_typeVegan
##                1.202904e+00                3.116012e-01
##         diet_typeVegetarian               home_typeFlat
##                1.626155e+00                5.755923e+00
##      home_typeSemi-detached           home_typeTerraced
##                3.028054e+00                3.363787e+00
##    uses_public_transportYes   annual_energy_consumption
##                6.106133e-01                1.000072e+00
##                      income              household_size
##                1.000001e+00                1.185329e+00
```

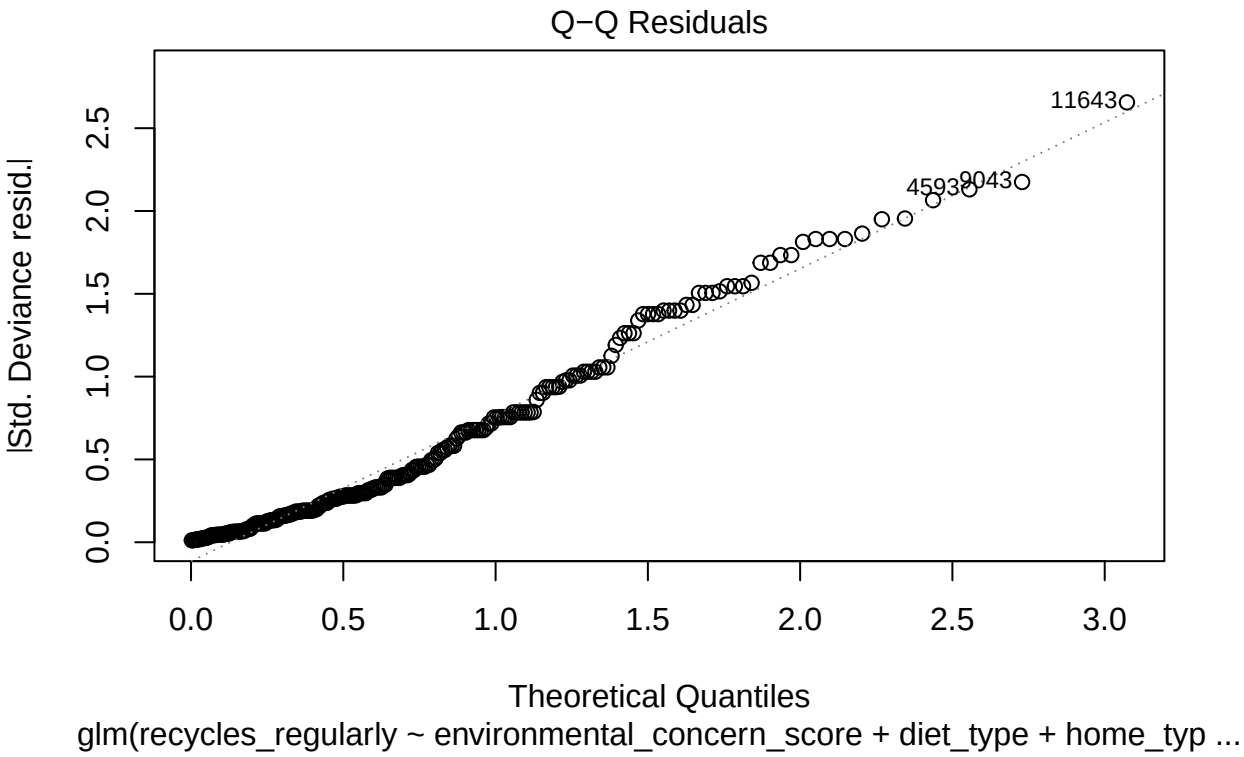
```
m5.glm <- glm( recycles_regularly ~ environmental_concern_score + diet_type + home_type, data = mydf, fa
summary(m5.glm)
```

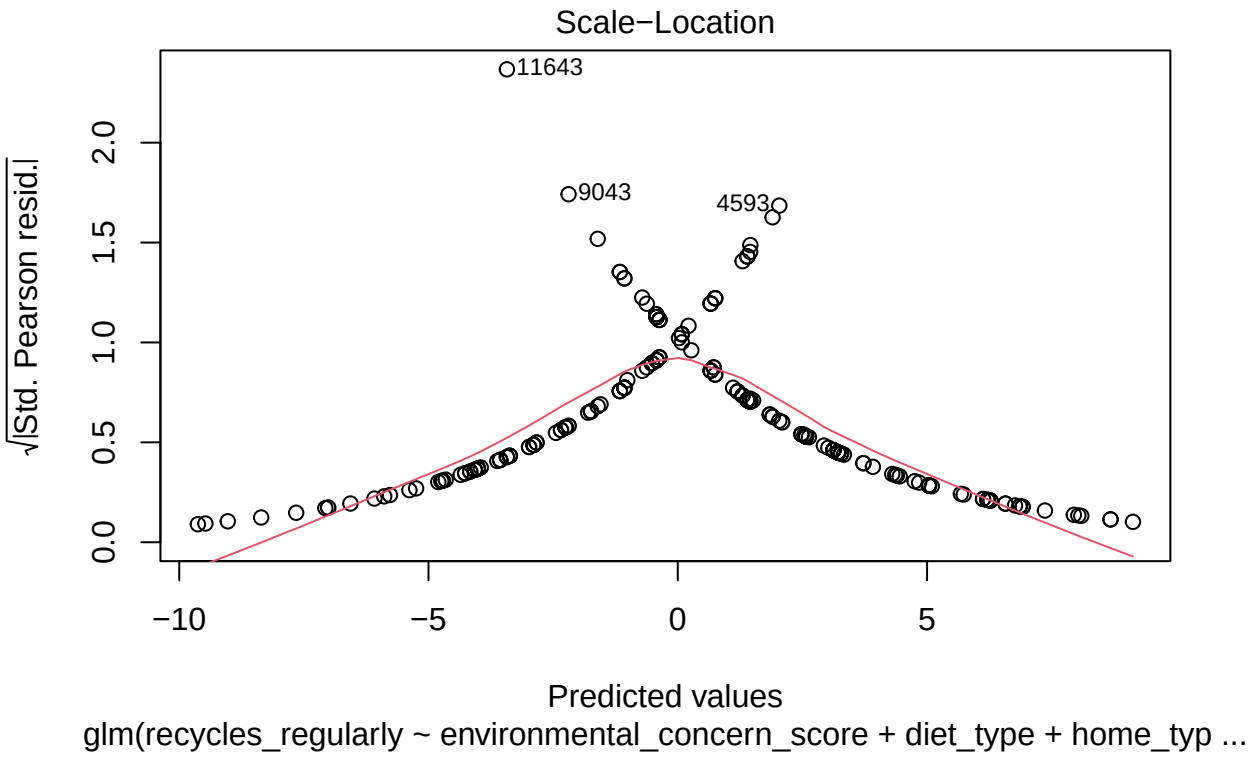
```
##
## Call:
## glm(formula = recycles_regularly ~ environmental_concern_score +
##     diet_type + home_type, family = binomial, data = mydf)
##
```

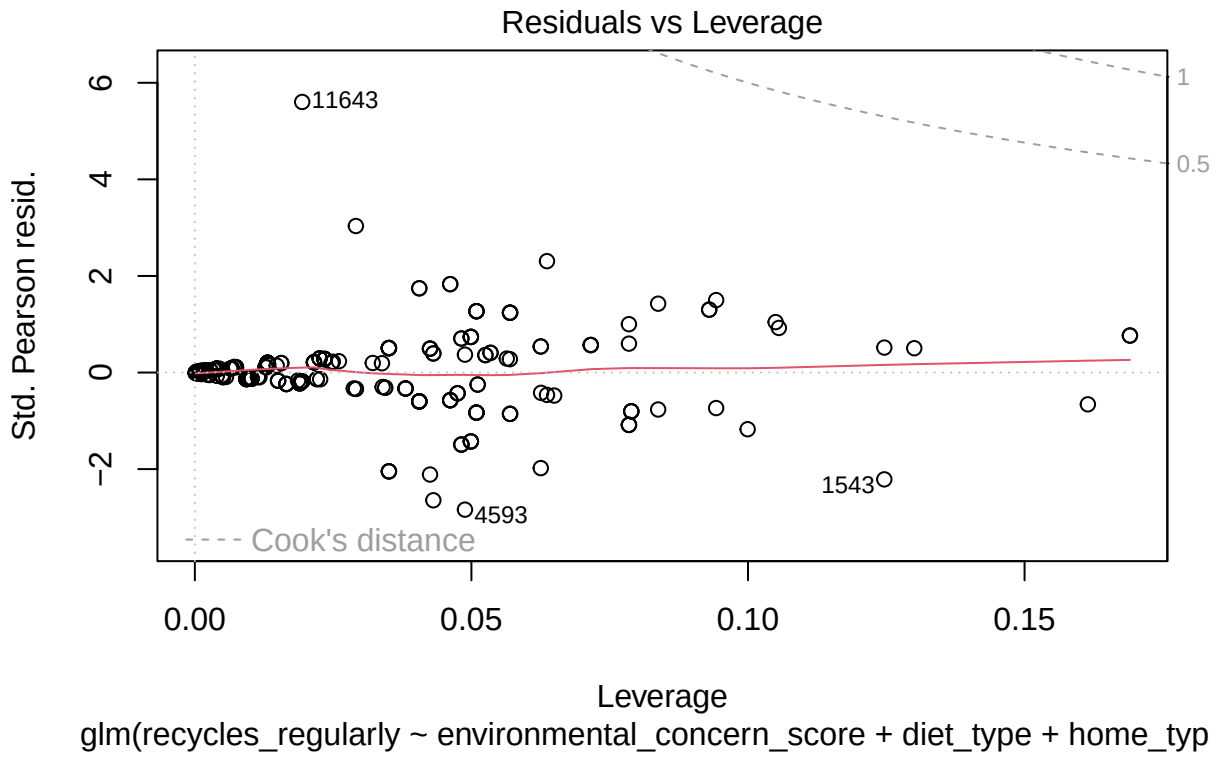
```
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  -11.29649    1.62810  -6.938 3.96e-12 ***
## environmental_concern_score    1.82123    0.25165   7.237 4.58e-13 ***
## diet_typePescatarian           0.05915    0.86714   0.068  0.94562
## diet_typeVegan                -1.18041    0.58758  -2.009  0.04454 *
## diet_typeVegetarian            0.44947    0.56998   0.789  0.43036
## home_typeFlat                  1.76509    0.64149   2.752  0.00593 **
## home_typeSemi-detached         1.02897    0.61962   1.661  0.09678 .
## home_typeTerraced              1.11920    0.62631   1.787  0.07394 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 326.33  on 235  degrees of freedom
## Residual deviance: 144.27  on 228  degrees of freedom
## AIC: 160.27
##
## Number of Fisher Scoring iterations: 6
```

```
plot(m5.glm)
```




Q−Q Residuals
|Std. Deviance resid.|
0.0
0.5
1.0
1.5
2.0
2.5
11643
45933
9043
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Theoretical Quantiles
glm(recycles_regularly ~ environmental_concern_score + diet_type + home_typ ...


Scale−Location
11643
9043
4593
√|Std. Pearson resid.|
0.0
0.5
1.0
1.5
2.0
−10
−5
0
5
Predicted values
glm(recycles_regularly ~ environmental_concern_score + diet_type + home_typ ...



###Stepwise model selection

```
m6.glm <- step(m5.glm)
```

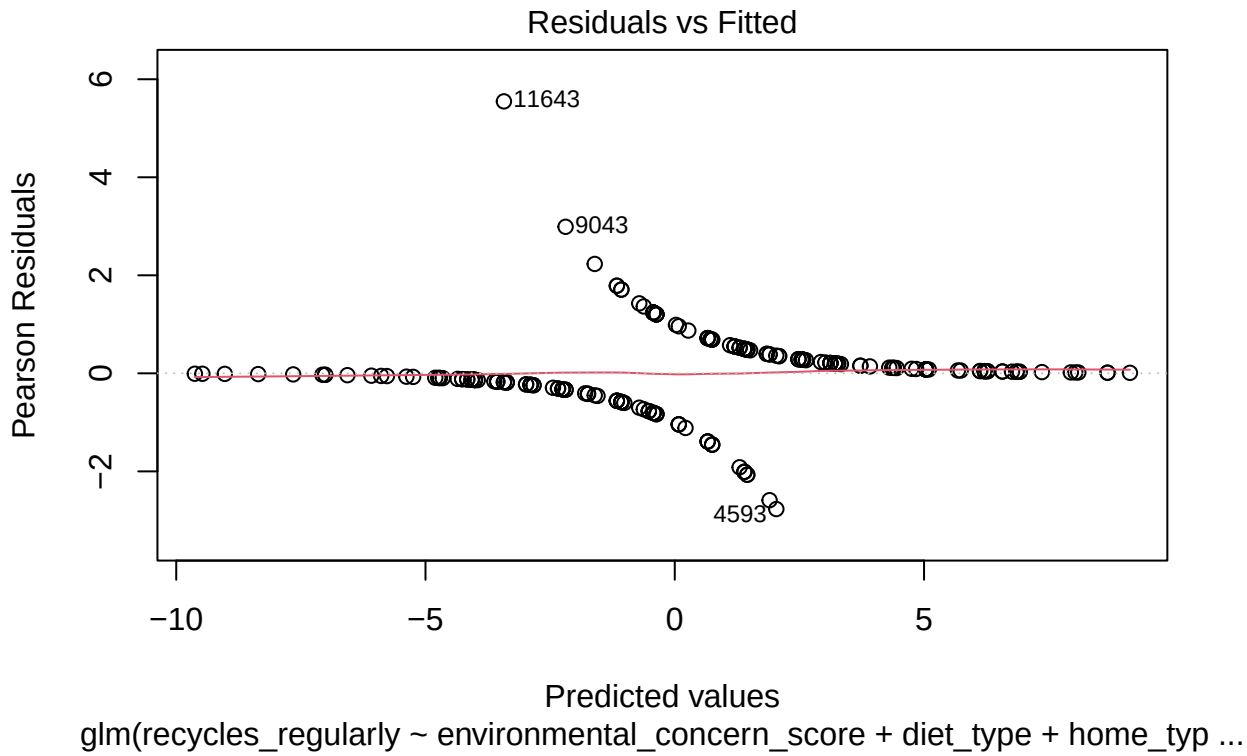
```
## Start:  AIC=160.27
## recycles_regularly ~ environmental_concern_score + diet_type +
##     home_type
##
##                               Df Deviance    AIC
## <none>                             144.27 160.27
## - diet_type                    3   150.38 160.38
## - home_type                    3   152.58 162.58
## - environmental_concern_score  1   317.28 331.28
```

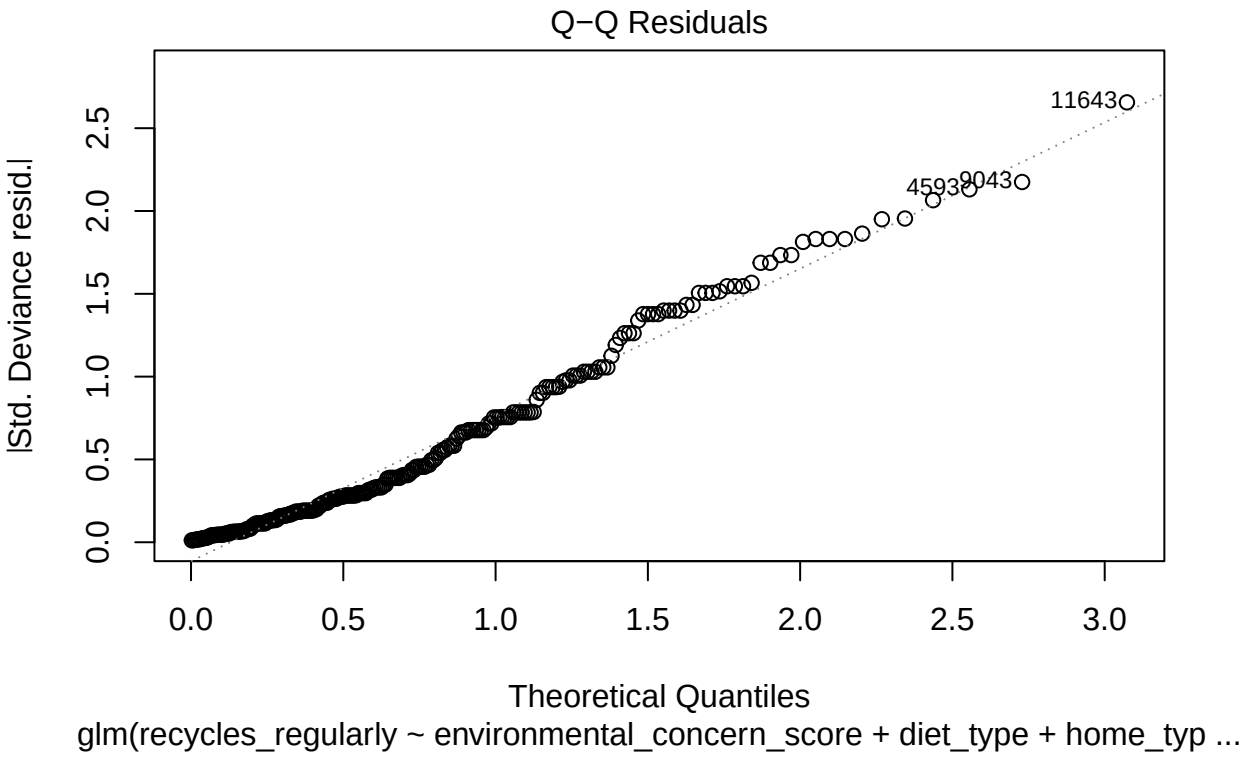
```
summary(m6.glm)
```

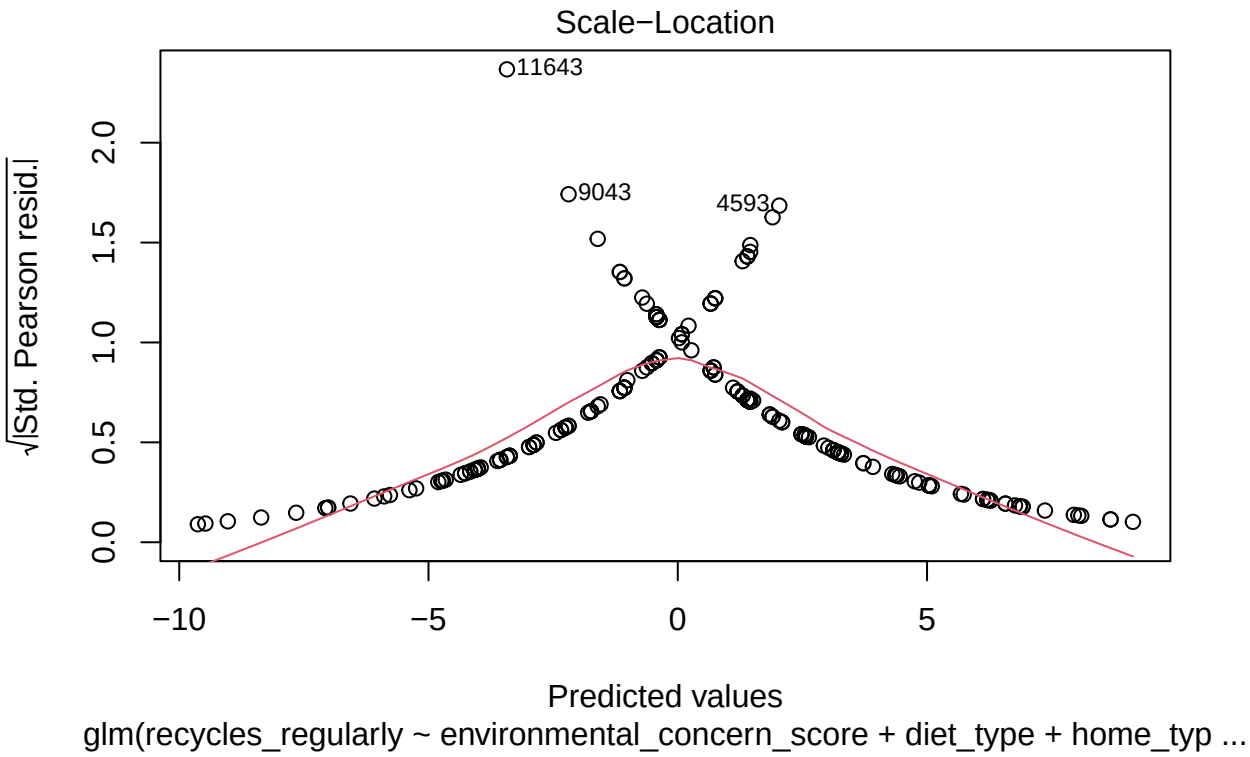
```
##
## Call:
## glm(formula = recycles_regularly ~ environmental_concern_score +
##     diet_type + home_type, family = binomial, data = mydf)
##
## Coefficients:
##                               Estimate Std. Error z value Pr(>|z|)
## (Intercept)                  -11.29649    1.62810  -6.938 3.96e-12 ***
## environmental_concern_score    1.82123    0.25165   7.237 4.58e-13 ***
```

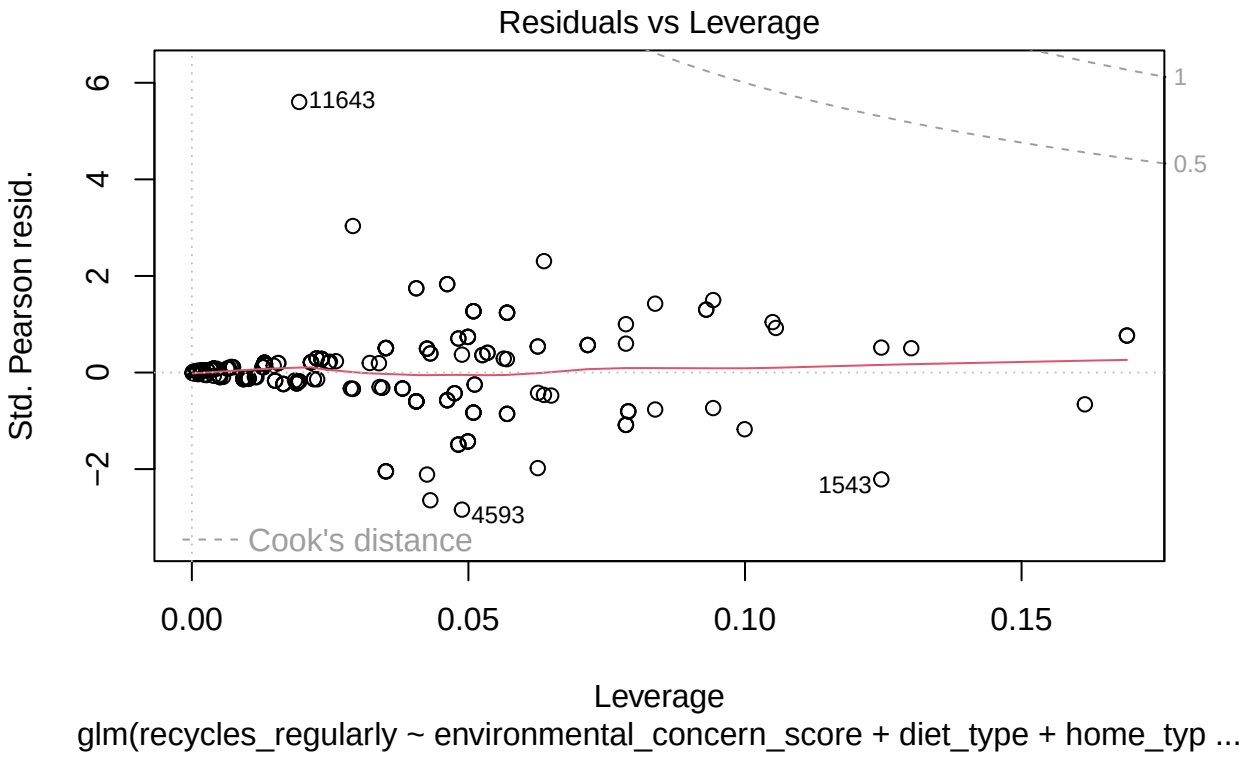
```
## diet_typePescatarian          0.05915    0.86714   0.068  0.94562
## diet_typeVegan               -1.18041    0.58758  -2.009  0.04454 *
## diet_typeVegetarian           0.44947    0.56998   0.789  0.43036
## home_typeFlat                 1.76509    0.64149   2.752  0.00593 **
## home_typeSemi-detached        1.02897    0.61962   1.661  0.09678 .
## home_typeTerraced             1.11920    0.62631   1.787  0.07394 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 326.33  on 235  degrees of freedom
## Residual deviance: 144.27  on 228  degrees of freedom
## AIC: 160.27
##
## Number of Fisher Scoring iterations: 6
```

```
plot(m6.glm)
```




Q−Q Residuals
|Std. Deviance resid.|
0.0
0.5
1.0
1.5
2.0
2.5
3.0
11643
45933
9043
Theoretical Quantiles
glm(recycles_regularly ~ environmental_concern_score + diet_type + home_typ ...


Scale−Location
11643
9043
4593
√|Std. Pearson resid.|
0.0
0.5
1.0
1.5
2.0
−10
−5
0
5
Predicted values
glm(recycles_regularly ~ environmental_concern_score + diet_type + home_typ ...


Residuals vs Leverage
Std. Pearson resid.
6
4
2
0
−2
11643
1543
4593
1
0.5
Cook's distance
0.00
0.05
0.10
0.15
Leverage
glm(recycles_regularly ~ environmental_concern_score + diet_type + home_typ ...